

Lectures on Radiotherapy Physics

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Lecture 1

Mathematical Background

“One reason why mathematics enjoys special esteem, above all other sciences, is that its propositions are absolutely certain and indisputable, while those of all other sciences are to some extent debatable and in constant danger of being overthrown by newly discovered facts. How can it be that mathematics, being after all a product of human thought which is independent of experience, is so admirably appropriate to the objects of reality?”

– Albert Einstein, 1921 address to the Prussian Academy of Sciences

“Whereof one cannot speak [clearly], thereof one must be silent.”

– Ludwig Wittgenstein, *Tractatus Logico-Philosophicus*

1.1 Connections Between Math and Physics

The legacy of the first modern physicists is not the notion that objects accelerate towards the Earth at the same rate regardless of their mass, or that the planets move around the sun in the shape of ellipses. Their largest contribution towards shaping modern technology and humanity’s understanding of the natural world, is showing that the laws of physics are formulated and tested under the purview of mathematics. The motion of the planets in a curved spacetime, the collisions between the elementary constituents of matter, the alignment of a subject’s atomic nuclei in an MRI machine, are all understood in the language of math.

It’s not by any means self-evident that the universe *has* to be this way – math as the language of physics, rather than tarot readings, astrological charts, or that we might have an unstructured universe with an absence of physical laws altogether. The connection between math and physics was mysterious even to Einstein – “How can it be that math, a product of human thought, is appropriate to the object of our experience?”

Mathematics is *logical truth* constructed entirely by human thought. Mathematical truths depend on the context that we give them. Using ordinary multiplication, the product $A \cdot B = B \cdot A$ if A and B are integers, but not necessarily if A and B are matrices, or rotations in 3 spatial dimensions, quantum mechanical operators, or other mathematical objects. The

quantity $2 \cdot 2 = 4$ in numerical base 10, but $2 \cdot 2 = 11$ in base 3. Modern mathematics always begins with a book of rules, a collection of **axioms**, for the logical system in question. Axioms are statements which are accepted to be true without proof. They form a ground level from which other properties of the system can be deduced with the rules of logic. The field of mathematics, for this reason, is characterized by rigor. Because it is built from the ground up, it is important that the foundations are defined very precisely.

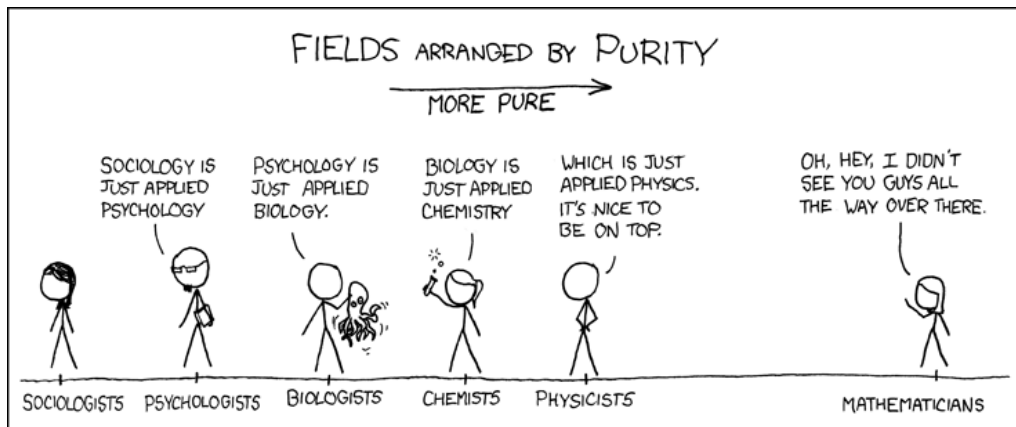


Figure 1.1: Munroe, R. "Purity." xkcd, no. 435, xkcd.com, 1 May 2008, <https://xkcd.com/435/>.

In physics, on the other hand, experiment always gets the final say. Experimental physics is concerned with *observational truths* according to the **scientific method** – we form hypotheses, collect, and analyze data. There are dozens of examples in the evolution of modern physics where the universe's behavior has defied physicists' intuition. Electromagnetic forces are mediated by ghostly "fields", light is always observed to propagate at the same speed, individual electrons propagate through a double slit and form wave-like patterns, like a buoys bobbing in water.

Theoretical physics is the human bridge between observation and understanding. How can it be that we measure light propagating at the same speed if I'm on Earth and you're on Alpha Centauri – a star system 40 trillion km away and moving at 20 km/s with respect to Earth? Over and over, surprising and seemingly nonsensical, but reproducible, measurements are shown to fit into the logical framework of mathematics. Given a new experimental result, *e.g.*, light propagating at a constant speed independent of relative motion, we bend our existing

laws and our understanding of the universe to try and explain these measurements in terms of new logically consistent principles. These new theories give us new falsifiable predictions, which we can then take to the lab to see if universe *really does* behave as the theory predicts.

We don't need very much math to go a long way in understanding physics. Einstein's Special Theory of Relativity, from which we deduce the famous $E = mc^2$, requires only high school algebra to arrive at its main results. Nevertheless, we do need some math to bridge the observation-understanding gap, even if it's only high school algebra, geometry, and a little bit beyond. That's not to say that I am claiming any pretense of mathematical rigor in these lectures. My hope for the year is to focus on concepts and telling a coherent story, but I don't want to shy away from a few math problems.

Every student of physics has, at one time or another, become frustrated with a computation. I want to take this opportunity at the beginning of the semester to remind you that this is part of the process. Anyone can learn physics with the right tools, the right teacher, and a bit of effort (see Figure 1.2).

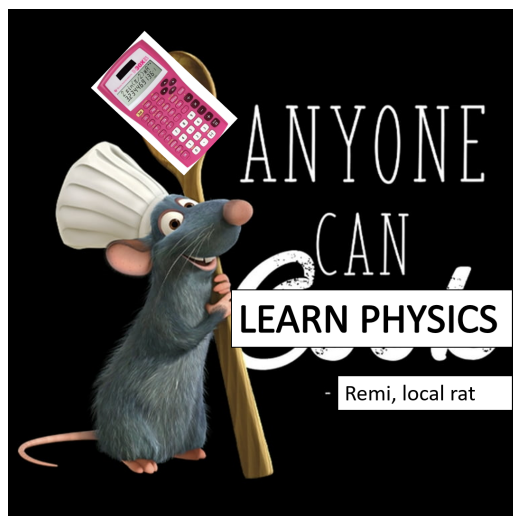


Figure 1.2: Anyone can learn physics!

I also want to share my perspective, that participating in understanding the world around us using the tools of mathematics is an understated opportunity. The fact that we're able to work out the motion of celestial bodies or calculate the probabilities of particles colliding

using nothing more than a blackboard and a scientific calculator, is dumbfounding. I hope that I can share some of my excitement about this material with you.

1.2 Mathematical Objects

1.2.1 Sets

At the bottom rung of the mathematical ladder are **sets**¹. For our purposes, you can think of sets as simply “collections” of mathematical objects. The objects belonging to a set are called its **elements**. The usual notation to indicate a set is curly brackets $\{\}$. Sets can include any object that we’re interested in studying mathematically. For example, we might have the set of outcomes from a coin flip $\{\text{Heads}, \text{Tails}\}$, or the set of integers $\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \pm 4, \dots\}$. The notation to indicate that an object is an element of a set is $\in$. Similarly, $\notin$ indicates that the object is not contained in the set.

$$-2, 1, 3000 \in \mathbb{Z} \tag{1.2.1}$$

$$\frac{2}{3}, \frac{1}{5}, \pi \notin \mathbb{Z} \tag{1.2.2}$$

For our purposes, we’ll mostly be concerned with a continuum of values, where any point on the number line is possible, out to arbitrarily many decimal places. This is called the set of **Real Numbers**, denoted by $\mathbb{R} = \{\text{all decimal expansions } \pm a_1 a_2 \dots a_n . b_1 b_2 \dots b_m\}$. Real numbers $a, b, c \in \mathbb{R}$ obey the familiar rules of arithmetic. For example,

$$a + b = b + a \tag{1.2.3}$$

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c \tag{1.2.4}$$

$$a(b + c) = ab + ac \tag{1.2.5}$$

...

1.2.2 Functions

As you might imagine, it is pretty uninteresting to have a collection of mathematical objects with which nothing happens. **Functions**, which I will interchangeably also call **maps**,

¹More precisely, Axiomatic Set Theory. I’ll be referencing so-called Naive Set Theory, which is overly precise for our purposes but not nearly precise enough for mathematicians.

are the mathematical machinery that takes inputs from one set and creates outputs in another set. When we talk about maps, it will be helpful to specify what types of inputs and outputs we're expecting. As an example, let's take the function f acting on natural numbers $x \in \mathbb{N}$, where $\mathbb{N} = \{1, 2, 3, 4, \dots\}$, mapping them to $\sqrt{x}$. The notation for this is to write

$$f : \mathbb{N} \rightarrow \mathbb{R} \quad (1.2.6)$$

$$x \mapsto \sqrt{x} \quad (1.2.7)$$

In this case, we've described a function which takes elements from $\{1, 2, 3, 4, \dots\}$ and maps them to $\{1, 1.41, 1.73, 2, \dots\}$. Here, $\mathbb{N}$ is called the **domain** of f , and $\mathbb{R}$ is called the **codomain**.

Although it's not always the case, many of the functions that we come across have well-defined **inverse functions**. For an invertible function f , its inverse f^{-1} , if it exists, has the property that

$$f^{-1}(f(x)) = x . \quad (1.2.8)$$

If we think of f as the machinery which turns x into a new variable, y

$$f(x) = y , \quad (1.2.9)$$

we can think of f^{-1} as the machinery which turns y back into x .

$$x \xrightarrow{f} y \xrightarrow{f^{-1}} x . \quad (1.2.10)$$

If we take any input x and apply f , we can apply f^{-1} to that output which will return our original input.

For example, take a function $f : \mathbb{R} \rightarrow \mathbb{R}$ which maps x to $f(x) = \frac{ax+b}{c}$, where a , b , and c are constants. This function has all of the nice properties that make it invertible. We can define a new variable y and solve for it.

$$\begin{aligned} f(x) = y &= \frac{ax+b}{c} \\ cy &= ax+b \\ cy-b &= ax \\ \frac{cy-b}{a} &= x \end{aligned}$$

We were given the machinery f that turns x in y from which we deduced the machinery which turns y into x ,

$$f^{-1}(y) = \frac{cy - b}{a} .$$

Note that the choice of the variable y is insignificant, and we could just as well represent the action of f^{-1} as

$$f^{-1}(x) = \frac{cx - b}{a}$$

or

$$f^{-1}(\xi) = \frac{c\xi - b}{a}$$

or

$$f^{-1}(\odot) = \frac{c\odot - b}{a} .$$

If we want to confirm that we have correctly computed the inverse, we can check that equation 1.2.8 holds: f^{-1} undoes whatever is done by f .

$$\begin{aligned} f^{-1}(f(x)) &= f^{-1}\left(\frac{ax+b}{c}\right) \\ &= \frac{c\left(\frac{ax+b}{c}\right) - b}{a} \\ &= \frac{ax+b-b}{a} \\ &= \frac{ax}{a} \\ &= x \end{aligned} \quad \checkmark$$

1.3 Euclidean Geometry

In 300 B.C.E., the Greek mathematician Euclid set out on the ambitious task of compiling everything that his contemporaries knew about mathematics. He called his treatise *Elements*, which amounted to 13 books at around 120,000 words. Among other things, *Elements* gives a formal description of the geometry of flat space beginning with a statement of 5 axioms. Hence, the subject of geometry in flat space, where the sum of a triangle's angles is always 180° and parallel lines remain parallel out to infinity, is called **Euclidean Geometry**.

1.3.1 Right Triangles

Properties of **right triangles** are useful since they will help us describe different distances in Euclidean Geometry. Right triangles are characterized by having one 90-degree angle. The longest side of the right triangle is called its **hypotenuse**. The lengths of the sides of all right triangles are related to each other through the Pythagorean Theorem. For a right triangle with sides of length a and b and hypotenuse of length c , the sides are related by

$$a^2 + b^2 = c^2 . \quad (1.3.1)$$

We will also need to understand the notion of **similar triangles**. Two triangles are called *similar* if they have identical shapes, but not necessarily the same size. Taking by fiat that all angles in a triangle add up to 180° , if two right triangles have one angle (other than the 90° angle) in common, their other angle ϕ is uniquely determined by $\phi[\text{degrees}] = 180^\circ - 90^\circ - \theta[\text{degrees}]$.

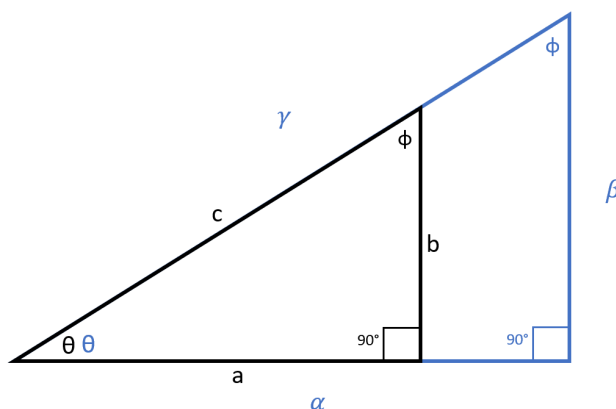


Figure 1.3: The black and blue triangles are similar to one another, meaning that they are identical in shape but have different sizes.

For two similar triangles as shown in Figure 1.3, the lengths of each side are in direct proportion with one another, or

$$\frac{a}{\alpha} = \frac{b}{\beta} = \frac{c}{\gamma} \quad (1.3.2)$$

These relationships will be useful when we discuss image magnification and how radiation field size changes with depth.

1.3.2 Trigonometry

Notice that because all right triangles with a common angle θ are similar, every angle from 0° to 360° defines an infinite set of similar right triangles (Figure 1.4).

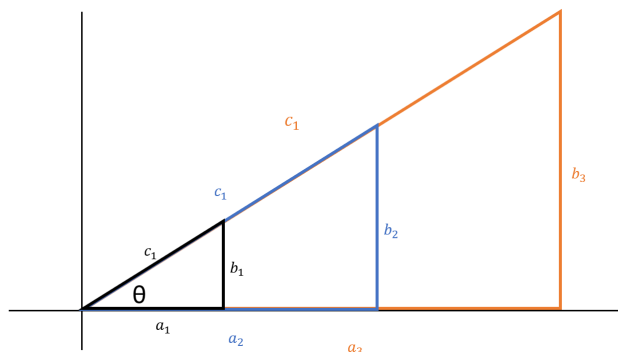


Figure 1.4: Every angle θ with respect to the x -axis describes an infinite set of similar right triangles.

All of these similar triangles have side lengths with the same ratios,

$$\frac{a_1}{c_1} = \frac{a_2}{c_2} = \frac{a_3}{c_3} = \dots \quad (1.3.3)$$

$$\frac{b_1}{c_1} = \frac{b_2}{c_2} = \frac{b_3}{c_3} = \dots \quad (1.3.4)$$

$$\frac{b_1}{a_1} = \frac{b_2}{a_2} = \frac{b_3}{a_3} = \dots \quad (1.3.5)$$

Since the ratios between any given pair of sides is always the same for a given θ , it is useful to define functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which map each angle to a ratio of side lengths. These are the sine, cosine, and tangent functions,

$$\sin(\theta) = \frac{a_i}{c_i} \quad (1.3.6)$$

$$\cos(\theta) = \frac{b_i}{c_i} \quad (1.3.7)$$

$$\tan(\theta) = \frac{b_i}{a_i} \quad (1.3.8)$$

where the index $i \in \mathbb{N}$ is just meant to indicate that we are taking side lengths from the same triangle as in 1.3.9-1.3.11. Given two out of three variables in any of the equations 1.3.9-1.3.11,

you can solve for the last value. A useful mnemonic for remembering which trig function to use is SOH-CAH-TOA.

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}} \quad (1.3.9)$$

$$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} \quad (1.3.10)$$

$$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}} \quad (1.3.11)$$

where “opposite” and “adjacent” are in reference to the angle θ . Each of these functions also admits an inverse,

$$\sin^{-1}(\sin(\theta)) = \theta \quad (1.3.12)$$

$$\cos^{-1}(\cos(\theta)) = \theta \quad (1.3.13)$$

$$\tan^{-1}(\tan(\theta)) = \theta \quad (1.3.14)$$

which can be used to calculate an angle given a ratio of right triangle side lengths.

1.3.3 Circles and Spheres

So far, we’ve only discussed angles as stated in degrees, where one degree is 1/360th of a complete rotation. An alternative definition is in terms of the arc length of a unit circle. Recall that all circles’ circumferences C are related to their diameters d by $C = \pi d$, or to their radii by

$$C = 2\pi r \quad (1.3.15)$$

Rather than degrees, it is common to define angles in terms of the arc length of the segment they carve out on a circle of radius $r = 1$. These units are called **radians**.

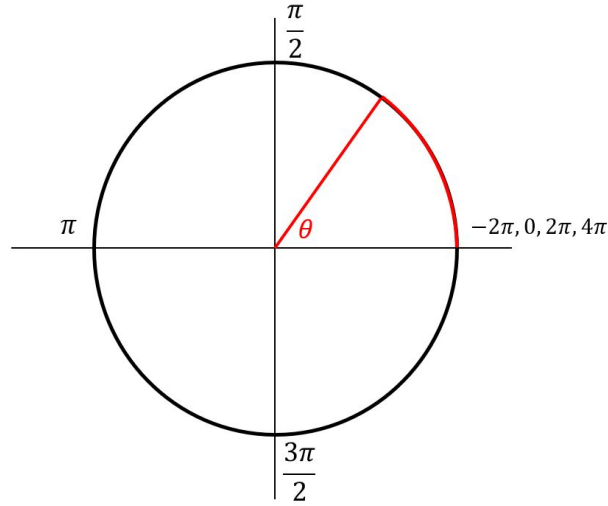


Figure 1.5: An angle θ with respect to the horizontal axis can be defined in terms of the arc length it carves out on a unit circle.

An angle of 0° carves out an arc of length 0, a 45° angle carves out a segment of length $\frac{\pi}{4}$, a 180° angle carves out π . To convert from degrees to radians, we can use

$$\theta[\text{radians}] = \theta[\text{degrees}] \left(\frac{2\pi}{360} \right) = \theta[\text{degrees}] \left(\frac{\pi}{180} \right) . \quad (1.3.16)$$

Remember also that the area A of a circle with radius r is given by

$$A(r) = \pi r^2 . \quad (1.3.17)$$

We will also need some properties of 3D spheres, namely their surface area

$$A(r) = 4\pi r^2 \quad (1.3.18)$$

and volume

$$V(r) = \frac{4}{3}\pi r^3 . \quad (1.3.19)$$

1.4 Scalars and Vectors

In section 1.2.2, we discussed functions with inputs and outputs of a single variable. In physics, these quantities are called **scalars**. Calling a value “scalar” just means it’s a solitary number – no direction associated, no nonsense. Examples of scalars include

- Mass
- Speed
- Distance
- Energy

When talking about physical systems, we're often concerned with functions involving spatial coordinates, moving particles, electromagnetic fields, etc., which require multiple numbers to fully describe. In these cases, the grouped collections of numbers are called **vectors**. Some vector quantities of interest include

- Position
- Velocity
- Momentum
- Force

For example, we might be interested in describing the position of a particle within 3 spatial dimensions. To do so, we might use an ordered triplet of real numbers representing its position along each of three orthogonal (forming right angles with one another) axes. These are called **Cartesian coordinates**. In 3 spatial dimensions, vectors are elements of the set $\mathbb{R} \times \mathbb{R} \times \mathbb{R} = \mathbb{R}^3$.

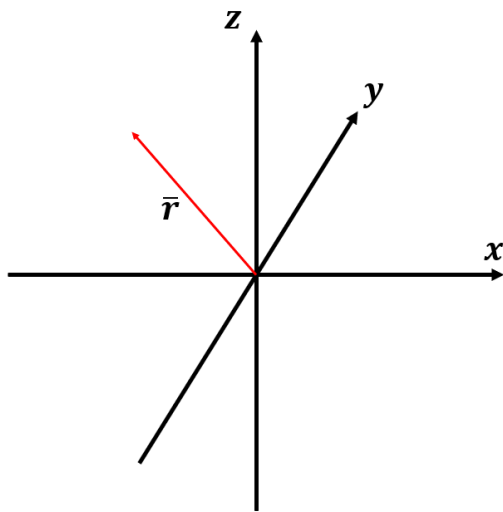


Figure 1.6: An arbitrary vector $\mathbf{r}$ represented in 3 Cartesian coordinates.

Because vector quantities are widely used in many different contexts, they are represented with a variety of notations. For a 3-dimensional position vector with displacement x in

the $\hat{\mathbf{x}}$ direction, displacement y in the $\hat{\mathbf{y}}$ direction, and displacement z in the $\hat{\mathbf{z}}$ direction, we might write the vector

$$\mathbf{r}(x, y, z) = \vec{\mathbf{r}}(x, y, z) \quad (1.4.1)$$

$$= \langle x, y, z \rangle \quad (1.4.2)$$

$$= x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}} \quad (1.4.3)$$

$$= \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (1.4.4)$$

all of which are equivalent.

A vector's length, or **magnitude**, is denoted by $\|\cdot\|$. In flat space, vector magnitude is calculated using the Pythagorean Theorem and positive square root,

$$\|\mathbf{r}(x, y, z)\| = +\sqrt{x^2 + y^2 + z^2} . \quad (1.4.5)$$

Notice that the vector magnitude is a scalar quantity.

1.4.1 Functions of Vectors

Functions can map vectors to other vectors, vectors to scalars, scalars to vectors, and scalars to scalars. The notation $f : A \rightarrow B$ is useful for clarifying the inputs and outputs of a function. For example, we might express the gravitational potential energy of a particle with mass m and height off of the ground z with the function $\phi : \mathbb{R}^3 \rightarrow \mathbb{R}$,

$$\phi(\mathbf{r}(x, y, z)) = mgz . \quad (1.4.6)$$

ϕ takes the object's coordinates $\langle x, y, z \rangle$ and spits out a scalar mgz .

As an example of a vector-to-vector function, we might have a magnetic field

$$\mathbf{B}(\mathbf{r}) = \langle B_x(\mathbf{r}), B_y(\mathbf{r}), B_z(\mathbf{r}) \rangle \quad (1.4.7)$$

as a function of position $\mathbf{r}(x, y, z)$. In this case, the independent components of the magnetic field vector depend on precisely where we are positioned in 3-dimensional space.

As an example of a scalar-to-vector map, we might describe the motion of a projectile under the influence of gravity as a function of time t , say $\mathbf{r}(t) = \langle x(t), y(t) \rangle$. If we take gravity to act in the $-\hat{\mathbf{y}}$ direction, the object's motion could be described by

$$\mathbf{r}(t) = \begin{bmatrix} v_{x,0}t + x_0 \\ -\frac{1}{2}gt^2 + v_{y,0}t + y_0 \end{bmatrix} \quad (1.4.8)$$

where $x_0 = x(0)$ and $y_0 = y(0)$ are initial positions of the object in the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ directions, and $v_{x,0}$ and $v_{y,0}$ are the initial speeds of the object in the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ directions, respectively.

1.5 Differential Calculus

First of all, calculus is not a prerequisite for this course, and you will not be expected to perform derivatives or integrals on your assignments. If you want to skip over this section, it will not be reflected in your grade. However, it's hard to exclude the discussion of calculus from our pedagogy entirely since it plays an important role in describing so many physical systems. I would like to put together the best narrative I can for the course, which I think includes a little bit of calculus.

Differential calculus is the study of instantaneous rates of change. For example, we might have a function $x(t)$ describing the position of my car as I drive from Austin to San Antonio, a distance of, say, 100 miles. If I leave Austin at 1 pm and arrive in San Antonio at 2 pm, we can infer that my average speed was $\frac{100 \text{ miles}}{1 \text{ hour}} = 100 \text{ mph}$. Although we know my average speed, we know nothing about how fast I was driving at any point during my trip. At different points during my journey, I might have been traveling at 50 mph, 80 mph, maybe even 200 mph. If we know $x(t)$, differential calculus gives us the tools to evaluate my speed at any point during my journey, called the **derivative** of x with respect to t .

Recall the function which maps out a line in the plane. Using our notation from section 1.2,

$$\begin{aligned} f : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto mx + b \end{aligned}$$

where m is the line's slope and $(0, b)$ is where f crosses the y-axis. For this function, the rate of change of f with respect to x is simple. If we take any two points x_1 and x_2 in the domain,

f grows by precisely the “rise over run”,

$$\begin{aligned}
 \frac{\Delta f}{\Delta x} &= \frac{f(x_2) - f(x_1)}{x_2 - x_1} \\
 &= \frac{mx_2 + b - (mx_1 + b)}{x_2 - x_1} \\
 &= m \frac{x_2 - x_1}{x_2 - x_1} \\
 &= m
 \end{aligned}
 \tag{1.5.1}$$

between those points. This should be unsurprising.

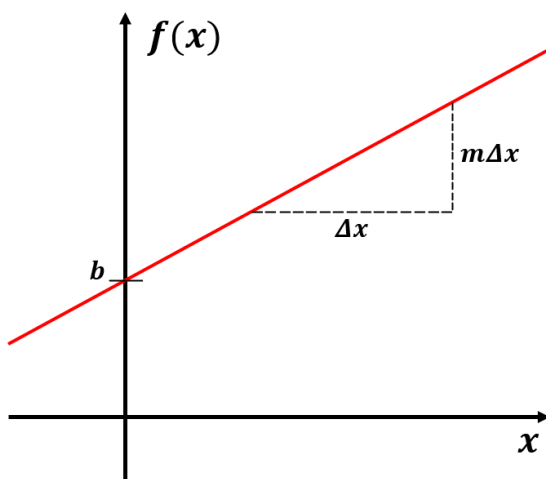


Figure 1.7: Plot of $f(x) = mx + b$ with positive m .

In single-variable differential calculus, the notion of finding the rate of change of f with respect to x is generalized to more complicated functions, where the slope is not constant over the entire domain.

We ask, for “well-behaved” functions $f : \mathbb{R} \rightarrow \mathbb{R}$, what is the slope of a linear approximation (a tangent) of f at every point. To calculate the derivative of an arbitrary function, we take the same expression for “rise over run” and let Δx get smaller and smaller, until it is *infinitesimal*.

Take, for example, the function $f(x) = x^3$. Say that we want to approximate the rate of change of f with respect to x at the point $x_0 = 2$. We might do so by taking points on

either side of x_0 , say 2 ± 2 , and evaluate $\Delta f/\Delta x$ like so,

$$\begin{aligned}\left(\frac{\Delta f}{\Delta x}\right)_{x_0 \pm 2} &= \frac{f(4) - f(0)}{4 - 0} \\ &= \frac{64 - 0}{4} \\ &= 16\end{aligned}\tag{1.5.2}$$

This is like how we approximated my speed while traveling from Austin to San Antonio – we took the distance between the two endpoints and divided by the total duration. The approximation is alright, but we’re still painting with broad strokes. If we wanted a closer approximation, we could use points which are closer to x_0 , say $x_0 \pm 1$.

$$\begin{aligned}\left(\frac{\Delta f}{\Delta x}\right)_{x_0 \pm 1} &= \frac{f(3) - f(1)}{3 - 1} \\ &= \frac{27 - 1}{2} \\ &= 13\end{aligned}\tag{1.5.3}$$

This is surely closer to the true rate of change of f with respect to x at $x_0 = 2$, but can we do better? In calculus, we take the **limit** of this expression for $\Delta f/\Delta x$ as $\Delta x \rightarrow 0$. Our macroscopic Δ ’s are replaced with microscopic d ’s to signify that we have done something novel.

$$\begin{aligned}f'(x) &= \frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{(x+h) - x} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} 3x^2 + 3xh + h^2 \\ &= 3x^2\end{aligned}\tag{1.5.4}$$

The above expression claims that for our function $f(x) = x^3$, f is changing by $3x^2$ per x at every point. If a particle’s position is described by the function $x(t) = t^3$, its speed, the rate of change of position with respect to time, is given by $s(t) = f'(t) = 3t^2$ at any instant.

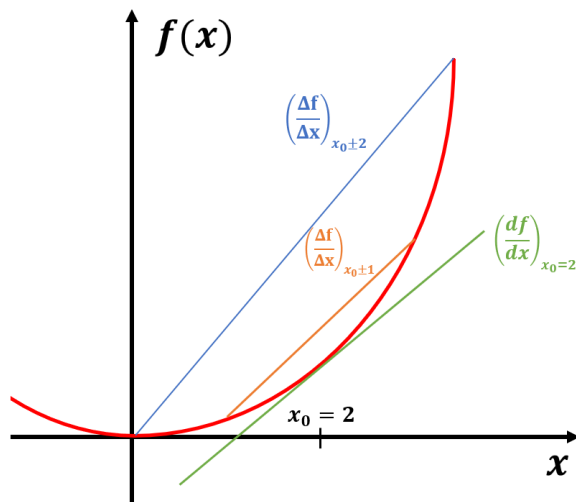


Figure 1.8: The derivative of f at point x_0 gives the slope of the tangent line passing through x_0 .

Many natural laws are expressed as **differential equations**. Differential equations characterize how quantities change with respect to one another. We will examine a few differential equations briefly. Again, I don't expect you to be able to solve them.

As an example of a differential equation, see the equation of motion for a mass m at position x on a spring where the mass is in equilibrium at $x = 0$, reproduced below,

$$m \frac{d^2 x}{dt^2} = -kx . \quad (1.5.5)$$

It shouldn't be that scary. All this equation says is that the acceleration of the mass at any moment as it bounces on the spring is directly proportional to how far it is pulled away from its equilibrium position (kx) and inversely proportional to its inertia (m).

1.6 Exponents and Logarithms

1.6.1 Exponents

Many phenomena in the natural world are described by exponential functions. When multiplying a number a by itself n times, we use the notation a^n where n is called the **exponent**, or **power**, of a

$$a^n = a \times a \times \cdots \times a \quad (n \text{ times}) . \quad (1.6.1)$$

It's a simple exercise to show the following properties of real numbers $a, b \in \mathbb{R}$ raised to exponents $n, m \in \mathbb{R}$.

$$\begin{aligned} \bullet \quad a^n \cdot a^m &= a^{n+m} & \bullet \quad (a^n)^m &= a^{nm} \\ \bullet \quad a^n \cdot b^n &= (ab)^n & \bullet \quad \left(\frac{a}{b}\right)^n &= \frac{a^n}{b^n} \end{aligned}$$

To show how to multiply negative exponents, take

$$\begin{aligned} \frac{a^n}{a^m} &= \frac{a \times a \times \cdots \times a \quad (n \text{ times})}{a \times a \times \cdots \times a \quad (m \text{ times})} \\ &= a^{n-m} = a^n \cdot a^{-m} \\ \rightarrow a^{-m} &= \frac{1}{a^m} \end{aligned} \tag{1.6.2}$$

Similarly, for fractional exponents, take the example

$$(a^n)^m = a \tag{1.6.3}$$

For this to be the case, we must have

$$\begin{aligned} (a^n)^m &= a^{nm} = a^1 \\ \implies m &= \frac{1}{n} \end{aligned}$$

so

$$a^{1/m} = \sqrt[m]{a} \tag{1.6.4}$$

1.6.2 Logarithms

Logarithms are defined as the inverse of exponential functions. If we have $f(x) = a^x$, then $f^{-1}(x) = \log_a(x)$ where a is called the **base** of the logarithm. Logarithms satisfy the quality

$$\log_a(a^x) = x \tag{1.6.5}$$

The properties of logarithms follow directly from the properties of exponents described above. Notably,

- $\log_a(x \cdot y) = \log_a(x) + \log_a(y)$
- $\log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y)$
- $\log_a(x^n) = n \log_a(x)$
- $\log_a(1) = 0$
- $a^{\log_a(x)}$
- $\log_a(x) = \frac{\log_b(x)}{\log_b(a)}$

We will need logarithms to describe properties of radioactive decay, radiation attenuation in vault shielding, and other natural phenomena.

1.6.3 Euler's Number

As we'll see, in discussions of exponential growth, the number $e = 2.71828\dots$ naturally arises time and time again. **Euler's number**, e , is a constant of nature which describes exponential growth in the same way that π is a constant of nature which relates the diameter of circles to their circumference.

To see how we arrive at $e = 2.71828\dots$, let's try to take some derivatives of exponential functions. If we have, for example, $f(x) = 2^x$, and we try to take the rise Δf over run Δx and let $\Delta x \rightarrow 0$,

$$\begin{aligned} \frac{d}{dx} \left(f(x) \right) &= \lim_{h \rightarrow 0} \frac{2^{x+h} - 2^x}{(x+h) - x} \\ &= \lim_{h \rightarrow 0} \frac{2^x 2^h - 2^x}{h} \\ &= 2^x \left[\lim_{h \rightarrow 0} \frac{2^h - 1}{h} \right]. \end{aligned} \tag{1.6.6}$$

Notice, first of all, that the quantity in brackets $[\cdot]$ is completely independent of x . In other words, the derivative $f'(x) = k \cdot f(x)$ where k is just a constant. If we plug in some small values of h (see Table 1.1, column 1), we find that the limit appears to converge to a value of about $k = 0.693$. Next, notice that the form of equation 1.6.7 has nothing to do with the number we chose to raise to the power x . If we have $f(x) = b^x$ for arbitrary real number b , we arrive at the same

$$\frac{df(x)}{dx} = b^x \left[\lim_{h \rightarrow 0} \frac{b^h - 1}{h} \right]. \tag{1.6.7}$$

If we plug in some values of b and h , we can observe how the limit in 1.6.7 changes (Table 1.1).

	b = 2	b = 2.5	b = 3	b = 4
h = 0.1	0.7177	0.9596	1.161	1.4870
h = 0.01	0.6956	0.9205	1.1047	1.3959
h = 0.001	0.6934	0.9167	1.0992	1.3873
h = 0.0001	0.6933	0.9163	1.0987	1.3864

Table 1.1: Various numerical values of equation 1.6.7, taking h to smaller and smaller values.

Notice in the bottom row of Table 1.1 that the limit in 1.6.7 passes from < 1 to > 1 . We might expect that this value varies continuously, where there must be a value c such that

$$\lim_{h \rightarrow 0} \frac{c^h - 1}{h} = 1 \quad (1.6.8)$$

This is exactly the case for $c = e = 2.71828\dots$

In other words, e is the unique number for which

$$\begin{aligned} \frac{d}{dx}(e^x) &= e^x \left[\lim_{h \rightarrow 0} \frac{e^h - 1}{h} \right] \\ &= e^x \end{aligned} \quad (1.6.9)$$

This is a property unique to the number e . If we have a function $f(x) = e^x$, we know that its derivative, second derivative, third derivative, and so on, all equal e^x .

Of note, we will often use the logarithm base- e , called a **natural logarithm**,

$$\log_e(x) = \ln(x) \quad (1.6.10)$$

$$\ln(e^x) = x \quad (1.6.11)$$

1.7 Dimensional Analysis

Dimensional analysis is a technique that will take you very far in physics. A variable's dimensions are the physical attributes that it is meant to represent, such as length L , mass M , time T , and temperature Θ . Square brackets are sometimes used to indicate a variable's dimensions. For example, if we write an expression for the kinetic energy of a particle of mass m and velocity v as

$$T = \frac{1}{2}mv^2, \quad (1.7.1)$$

we can write the dimensions of m as

$$[m] = M = \text{mass} , \quad (1.7.2)$$

or the dimensions of v as

$$[v] = \frac{L}{T} = LT^{-1} = \frac{\text{length}}{\text{time}} . \quad (1.7.3)$$

Of note, the quantity $\frac{1}{2}$ is dimensionless,

$$\left[\frac{1}{2}\right] = 1. \quad (1.7.4)$$

The dimensions of the energy T are just the product of the dimensions of $\frac{1}{2}mv^2$,

$$[T] = 1 \cdot M \cdot \frac{L}{T} \cdot \frac{L}{T} = ML^2T^{-2} \quad (1.7.5)$$

The idea behind dimensional analysis is simple – that the left and right sides of our equations must have the same dimensions in order for them to make any sense. We will disallow any equations which look like

$$\text{apples} = \text{oranges} \quad (1.7.6)$$

or

$$\text{apples} + \text{oranges} = \text{apples} . \quad (1.7.7)$$

The left- and right-hand sides of each equation, and each additive term within our equations, must have the same dimensions. For example, we can check that the expression for a projectile under the influence of no forces (the x value in equation 1.4.8) is dimensionally consistent.

$$\begin{aligned} [x(t)] &= [v_{0,x}t] + [x_0] \\ L &= \frac{L}{T} \cdot T + L \\ L &= L + L \quad \checkmark \end{aligned} \quad (1.7.8)$$

This is a useful sanity check to make sure we haven't made any egregious errors. Dimensional analysis is also useful for helping remember the dimensions of variables in our equations. For

example, if we can't remember the dimensions of the gravitational constant g , but we remember the expression for the position of a body in free fall $y(t) = -\frac{1}{2}gt^2$, we can solve for $[g]$.

$$\begin{aligned}[y(t)] &= [-\frac{1}{2}][g][t^2] \\ L &= 1 \cdot [g] \cdot T^2 \\ \rightarrow [g] &= LT^{-2}\end{aligned}$$

Notice that I am making a distinction between a quantity's dimensions and its **units**. In the expression above, the dimensions of $y(t)$ are always length L . However, we have a choice of which units we want to use to describe that length, e.g., mm, m, ft, miles, light-years, furlongs, etc.

We will frequently use the prefixes shown in Table 1.2 to scale our units to the appropriate order of magnitude. For example, 5 nanoseconds is equivalent to

$$5 \text{ ns} = 5 \cdot 10^{-9} \text{ s}$$

and π megavolts is

$$\pi \text{ MV} = \pi \cdot 10^6 \text{ V} . \tag{1.7.9}$$

milli (m)	10^{-3}	kilo (k)	10^3
micro (μ)	10^{-6}	mega (M)	10^6
nano (n)	10^{-9}	giga (G)	10^9
pico (p)	10^{-12}	tera (T)	10^{12}
femto (f)	10^{-15}	peta (P)	10^{15}

Table 1.2: Unit prefixes.

Lecture 2

Overview of Modern Physics

To talk about radioactive decay, radiation transport, magnetic resonance, and other topics in radiology and radiation oncology, we will require many concepts from many fields of physics, a body of knowledge built over about 350 years. Our approach will be to talk about physics historically from a bird's-eye-view, mostly in chronological order, making stops along the way to pick out results that are relevant for us.

2.1 Newtonian (Classical) Mechanics

Classical mechanics is a theory developed by Isaac Newton in the late 17th century. It aims to describe the motion macroscopic objects within our everyday lives, such as the motion of projectiles under the influence of gravity, rotating rigid bodies, properties of fluids, swinging pendulums, oscillating masses on springs, celestial orbits, propagating sound waves, atomic collisions, and many other phenomena. To its credit, the few principles on which Newtonian Mechanics is based go very far. Unfortunately, however, the Newtonian treatment of classical dynamics fails to describe our universe in the extreme regimes of the very fast, the very heavy, and the very small. To describe these realms in a way that, (1) agrees with experiment, and (2) reproduces the results of prior experiments in the appropriate limits (the speed of light $c \rightarrow \infty$, Planck's constant $h \rightarrow 0$). Namely, will we need the theories of relativity, quantum mechanics, and quantum field theory, which were not developed until the 20th century. Nevertheless, the Newtonian principles that at one time seemed fundamental, still adequately describe much of the familiar world.

Front and center in the Newtonian worldview are **forces**. Similar to our sets in the absence of functions as in section 1.2, objects without forces make for pretty unexciting physical systems. Forces are the physical impetus which allow material objects to “talk to each other.”

At Newton’s time, only two fundamental forces² had been discovered, gravity and electromagnetism. In the modern understanding of physics, there are two additional movers called the strong and weak nuclear forces. Because the strong and weak forces are only prominent over ranges on the order of the size of an atomic nucleus ($\approx 10^{-15}$ m), we can discuss Newtonian mechanics of without much regard for them.

2.1.1 Newton’s Laws

The postulates of classical mechanics from which dynamics are derived are called **Newton’s Laws**. Newton’s Laws describe how massive bodies act under the influence of forces. The three laws are

1. There exist coordinate systems for which objects in motion stay in motion unless acted upon by some force, $\mathbf{F} = 0 \rightarrow \mathbf{a} = 0$.
2. Under force $\mathbf{F}$, an object will accelerate $\mathbf{a}$ in proportion to its mass m , i.e., $\mathbf{F} = m\mathbf{a}$ (or more precisely, $\mathbf{F} = \frac{d\mathbf{p}}{dt}$).
3. Every action has an equal and opposite reaction. The force of object A acting on B is equal in magnitude and opposite in direction to the force of object B acting on A, $\mathbf{F}_{A,B} = -\mathbf{F}_{B,A}$.

From Newton’s laws, and a description of the forces at play, the time-evolution of the system follows. Take, for example, masses m_1 and m_2 under the influence of the force of gravity with magnitude

$$F_g = \frac{Gm_1m_2}{r^2} . \quad (2.1.1)$$

where r is the scalar distance between the masses. If we have the gravitational force F of the Earth of mass m_e acting on a baseball of mass m_b at a distance of y from the Earth’s center of mass, the baseball experiences a force

$$F = -\frac{Gm_em_b}{y^2} = m_b a_b . \quad (2.1.2)$$

²“Fundamental” meaning that they can’t be understood in terms of other phenomena at their roots. Friction, for example, arises as a consequence of electromagnetic interactions at the interface between two objects.

Recognizing the acceleration as the second derivative of time with respect to the baseball's position, this gives us a differential equation describing the motion of the ball,

$$m_b \frac{d^2 y}{dt^2} = -\frac{G m_e m_b}{y^2} . \quad (2.1.3)$$

This is the routine in Newtonian mechanics – write out the the forces in the problem, then turn the mathematical crank to describe the motion of the system. This formula works very well for most practical, everyday calculations. If we want to build a bridge, or a skyscraper, or an airplane, the magnitudes of the calculation errors using the Newtonian framework as opposed to relativistic or quantum mechanical frameworks are quite small. Even on the planetary scale, the path of mercury's orbit in the night sky, as calculated with Newtonian mechanics, differs from measurement and the value predicted by relativity by only 43 arcseconds/century = $\frac{43}{3600}$ degrees/century.

For convenience, we often calculate forces using the S.I. unit called Newtons, where

$$1 \text{ N} = 1 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \quad (2.1.4)$$

This is the force “felt” by a 1 kg mass on a planet with a gravitational acceleration $g = 1 \text{ ms}^{-2}$

2.1.2 Conserved Quantities

Although this notion of balancing forces and calculating a system's dynamics gets left by the wayside in later theories, there are some principles that follow from the Newtonian formulation which we will take with us, namely, the conservation of energy, momentum, and angular momentum.

Energy, in physics, represents an entity's capacity to do work or to create change within its environment. We express the energy of objects as the sum of their kinetic energy and potential energy. In classical mechanics, the **kinetic energy** T of a mass m at position $\mathbf{x}(t)$ is given by

$$T = \frac{1}{2} m v^2 = \frac{1}{2} m \left(\frac{d\mathbf{x}}{dt} \right)^2 . \quad (2.1.5)$$

Hopefully it's somewhat intuitive that objects in motion have some capacity to put other objects in motion. I wouldn't dwell too much on the mathematical form of the kinetic energy

here with the square of the velocity; kinetic energy at this point in history was simply known to be “the thing that is conserved in collisions”. James Prescott Joule, the discoverer of energy as a conserved quantity and the namesake of the S.I. unit of energy, would have referred to kinetic energy as *vis viva*, the “living force”. The unit 1 Joule is defined

$$1 \text{ J} = 1 \frac{\text{kg m}^2}{\text{s}^2} \quad (2.1.6)$$

Potential energy, on the other hand, characterizes work that has yet to be done by a force. A ball sitting at the top of a hill or two positive electric charges sitting next to each other have some stored energy from which motion might arise – given the opportunity, the ball will roll down hill and the positive charges will push each other away. We often use the characters U or V to represent potential energy in a given system. For example, a particle of mass m on Earth sitting at height h off of the ground might be said to have potential energy

$$U(h) = mgh \quad (2.1.7)$$

where $g = (\frac{GM_E}{R_E^2}) = 9.8 \text{ ms}^{-2}$ is the gravitational constant here on Earth. U , again, represents the untapped *energy* stored within the gravitational field of Earth and is embodied by the *force* of gravitation attraction. The mathematical relationship between potential energies and forces they create³ is

$$\mathbf{F} = -\frac{dU}{d\mathbf{x}} \quad (2.1.8)$$

In any closed physical system with masses m_i at positions $\mathbf{x}_i$, the total energy within the system is equal to the sum of the energies of each mass,

$$E = \sum_i T_i + \sum_i U_i \quad (2.1.9)$$

$$= \sum_i \left(\frac{1}{2} m_i \left(\frac{d\mathbf{x}_i}{dt} \right)^2 + U(x_i) \right) \quad (2.1.10)$$

³Strictly speaking, equations 2.1.9, 2.1.11, and 2.1.8 describe *conservative* forces such as gravity and electromagnetism. If we allow some other, non-fundamental forces such as friction, drag, and viscosity into our system, these equations don’t necessarily hold. The same conservation laws are still broadly true, but our “system” must be expanded to include those energy sinks, for example, thermal energy lost to friction.

This quantity is always **conserved**, meaning it is unchanging in time.

$$\frac{dE}{dt} = 0 \quad (2.1.11)$$

Often in Newtonian Mechanics, the conservation of energy, momentum, and angular momentum, helps us simplify problems. For example, if we have a pizza at the top of the Leaning Tower of Pisa under the influence of gravity, it “wants” to drop to the ground, minimizing its potential energy, per equation 2.1.8. Assuming it does so without energy loss to drag, its potential energy is perfectly converted to kinetic energy. We can express the particle’s velocity at any point during the drop

$$mg(h_0 - h) = \frac{1}{2}mv^2 \quad (2.1.12)$$

$$\rightarrow v(h) = \sqrt{2g(h_0 - h)} \quad (2.1.13)$$

This energetic approach gives us an alternative to balancing forces and solving the resulting differential equations of motion. In later theories, and non-Newtonian formulations of classical mechanics, this tendency of nature to favor less energetic states is treated as more fundamental, being the driver of change.

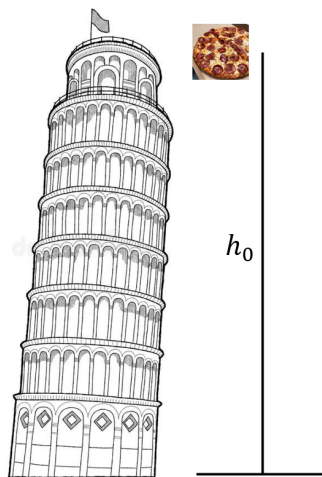


Figure 2.1: Neglecting drag, a pizza of mass m dropped from a height h_0 will have velocity $v(h) = \sqrt{2g(h_0 - h)}$ during its descent. Potential energy mgh stored in the gravitational field is converted to kinetic energy of the pizza.

Another quantity which is conserved in closed systems is momentum $\mathbf{p} \in \mathbb{R}^3$. The momentum of a mass m_i is defined as the product of its mass and velocity,

$$\mathbf{p}_i = m_i \frac{d\mathbf{x}_i}{dt} = m\mathbf{v}_i \quad (2.1.14)$$

Similar to energy, we always find that the sum of momenta in any closed system is unchanging in time

$$\sum_i \frac{d\mathbf{p}_i}{dt} = \mathbf{0} . \quad (2.1.15)$$

This is another useful concept for calculating dynamics of something like collisions. If I have a tennis ball and a bowling ball colliding into each other, I can work out the dynamics of the collision using conservation principles, without ever invoking Newton’s Laws.

Lastly, in systems with rotating bodies, a quantity called angular momentum

$$\mathbf{L} = \mathbf{x} \times \mathbf{p} = \mathbf{x} \times m \frac{d\mathbf{x}}{dt} \quad (2.1.16)$$

is also conserved. In equation 2.1.16, the symbol $\times$ represents an operation called the vector cross product. We won’t be performing any calculations using “ $\times$ ”, but we’ll get a feel for what it means in the next section on electromagnetism. At a glance, the cross product between two vectors is proportional to the magnitude of the vectors $||\mathbf{L}|| \propto ||\mathbf{x}|| \cdot ||\mathbf{p}||$, but points in a direction which is perpendicular to both $\mathbf{x}$ and $\mathbf{p}$.

You might ask, why should any of these quantities be conserved in the first place? Indeed, conservation of energy, conservation of momentum, and conservation of angular momentum, all have deep roots which are more pleasing than just being random quantities which happen to be constant in time. We won’t prove it here, but these laws all directly follow from different symmetries that are inherent within the laws of physics. This realization, that conservation laws follow from invariances, is called **Noether’s Theorem**, after mathematician Emmy Noether. Conceptually, Noether’s theorem demonstrates that

- energy conservation follows from time translation invariance – the laws of physics don’t care what time we call $t_0 = 0$,
- momentum conservation follows from space translation invariance – the laws of physics don’t care which position we call $\mathbf{x} = \mathbf{0}$,

- angular momentum invariance follows from rotation invariance – the laws of physics don’t care whether we rotate all of our coordinates by θ degrees.

2.2 Classical Electromagnetism

Documentation of electric and magnetic phenomena dates back to antiquity. Ancient Greek philosopher Thales of Miletus observed that amber, when rubbed against wool, could attract light objects such as feathers. Hence⁴, the English word electricity is derived from the Greek $\etaλεκτρον$, meaning “amber.” Similarly, he documented the attraction of iron and lodestones, naturally occurring permanent magnetic stones which are now believed to have acquired their magnetization from lightning strikes.

Scientific understanding of electromagnetism did not progress significantly until the 1700’s, when Charles Augustin de Coulomb experimentally observed a mutual attraction between opposite electric charges, and repulsion of alike electric charges, with a force of

$$\mathbf{F}_E = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{\mathbf{r}} \quad (2.2.1)$$

where ϵ_0 is a constant, q ’s are the charge magnitudes, and r is the distance between the charges. Here, a vector of magnitude $||\hat{\mathbf{r}}|| = 1$ is used to indicate that the direction of the force is parallel to a line drawn between q_1 and q_2 .

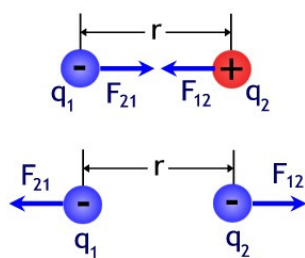


Figure 2.2: Opposite charges attract, and like charges repel, in proportion to the magnitude of the charges and the inverse square of the distance between them. http://ffden-2.phys.uaf.edu/212_fall2003.web.dir/don_bahls/coulombs_law.html

⁴Sorry for saying hence.

It's worth mentioning that a common unit of electric charge is the Coulomb C , although a more natural unit is the charge of a single electron, the elementary charge e . The two are related by

$$1\text{ e} = -1.602 \times 10^{-19}\text{ C} . \quad (2.2.2)$$

The minus sign convention, having been passed down for two centuries, is due to an arbitrary assignment by Benjamin Franklin when performing an experiment rubbing glass rods with silk.

2.2.1 Fields and Forces

Importantly, from the study of electricity and magnetism came a theory of fields. Fields are physical entities which permeate all of space, acting upon bodies which cohabitate with them. An often used visualization for electromagnetic fields are field lines, shown in Figure 2.3, whose density signifies the relative strength of the field at each point – right next to the charges the field lines group up, but far away from them they spread out.

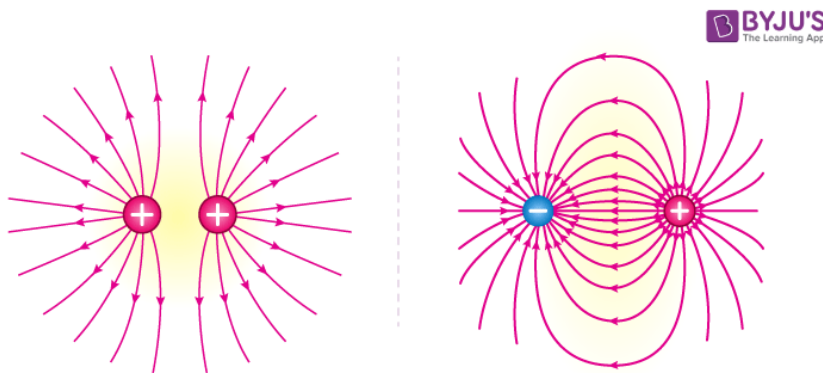


Figure 2.3: EM fields placeholder. <https://byjus.com/physics/electric-field-lines/>

Equation 2.2.1 above, can be restated in terms of an electric field due to q_1 acting upon q_2 .

$$\mathbf{F}_E = q_2 \mathbf{E}_1 = q_2 \left(\frac{1}{4\pi\epsilon_0} \frac{q_1}{r^2} \hat{\mathbf{r}} \right) \quad (2.2.3)$$

The physical interpretation emphasized here is that electric charge q_1 emits an electric field, a function $\mathbf{E}(\mathbf{r}) : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, everywhere in space around it. That field then kicks q_2 in proportion to the magnitude $\|\mathbf{E}\|$, decreasing in strength as $\sim \frac{1}{r^2}$. Although we haven't explained how

the electric field propagates, we have made the distinction that the charges q_1 and q_2 , which may be on opposite ends of the universe, don't interact with each other directly. The *field mediates the interaction between charges*.

Magnetic fields and forces arise from electric charges in motion. A charge q moving with velocity v creates a magnetic field around it with strength

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{qv}{r^2} (\hat{\mathbf{v}} \times \hat{\mathbf{r}}) . \quad (2.2.4)$$

where μ_0 is just a constant. Similar to equation 2.2.3, the magnitude of B falls off as $\sim \frac{1}{r^2}$. It is proportional to q and the magnitude of the velocity v . The direction of $\mathbf{B}$ is indicated by the cross product $\hat{\mathbf{v}} \times \hat{\mathbf{r}}$; this is a vector whose direction forms right angles with both the particle's velocity and position.

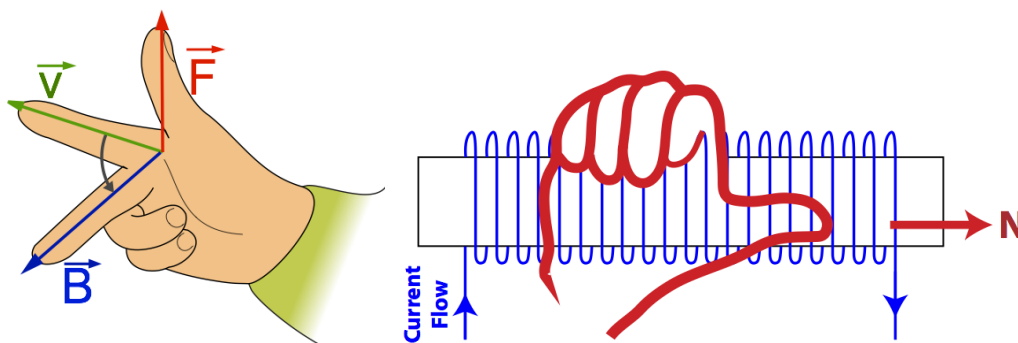


Figure 2.4: The direction of a magnetic field due to a current, and direction of force on a particle moving in a magnetic field, can be determined with the right hand rule. <https://www.aplusphysics.com/courses/honors/magnets/electromagnetism.html>

The motion of a charged particle under the influence of a magnetic field is equally strange. In the presence of a magnetic field without an electric field, the force felt by a particle of charge q and velocity $\mathbf{v}$ is

$$\mathbf{F}_B = q\mathbf{v} \times \mathbf{B} . \quad (2.2.5)$$

If we have both electric and magnetic fields present, the particle is directed according to the **Lorentz Force Law**,

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) . \quad (2.2.6)$$

There's a lot to unpack in equations. For now, we'll just keep moving, having established

1. Static electric charges emit electric fields, permeating the space around them.
2. Moving electric charges emit magnetic fields, permeating the space around them.
3. Electric and magnetic fields act on charges, mediating interactions between them.

We will circle back around to talking about right hand rules and motion of charged particles in electromagnetic fields in later lectures.

2.2.2 Electromagnetic Waves

Properties of electric and magnetic fields were developed as a product of empirical observations over the course of the 19th century. The theory of classical electrodynamics culminated in the statement of four equations, collectively called **Maxwell's Equations**, which describe

- $\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho_e$: how electric fields are generated by electric charges
- $\nabla \cdot \mathbf{B} = 0$: the absence of magnetic charges, called “magnetic monopoles”
- $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$: how changing magnetic fields create electric fields
- $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$: how changing electric fields create magnetic fields

Here, ϵ_0 and μ_0 are just constants, and the vector $\mathbf{J}$ represents an electric current⁵. James Clerk Maxwell, the namesake of these equations, is really only responsible for adding one term – the $\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$ at the end of the last equation. With this term, however, comes very important implications.

The last two equations, with Maxwell's addition, tell us that changing electric fields over time generate magnetic fields, and changing magnetic fields over time generate electric fields. These equations, when combined, have solutions which take the form of propagating **electromagnetic (EM) waves**. Waves are omnipresent in physics. Ripples in water, vibrating guitar strings, seismic activity, and propagating sound all have similar mathematical descriptions. With Maxwell's discovery, EM waves were added to the list of wave phenomena.

⁵More precisely, an electric current *density* with dimensions of charge per area per time.

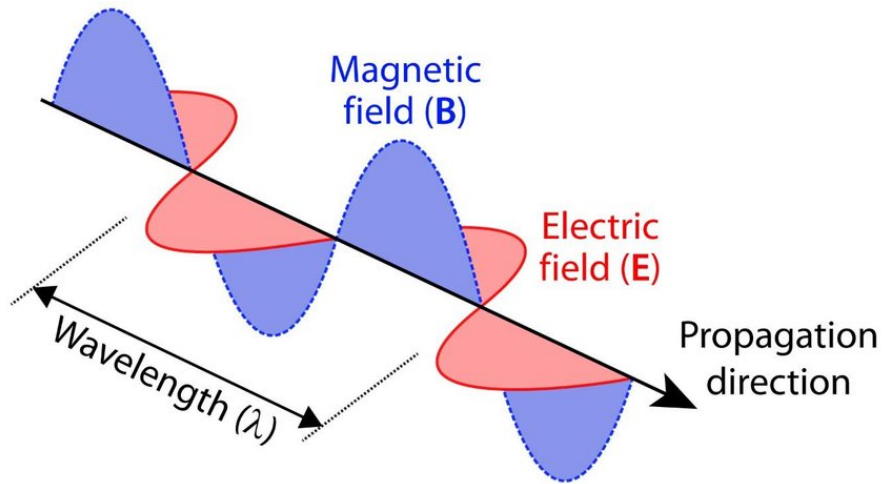


Figure 2.5: Maxwell's equations *in vacuo* have wave solutions with perpendicular electric and magnetic components. https://www.researchgate.net/figure/An-electromagnetic-wave-consisting-of-electric-and-magnetic-oscillating-fields-In-this_fig1_320616988

The speed at which electromagnetic waves propagate was first calculated as a combination of constants ϵ_0 and μ_0 from the study of electric and magnetic phenomena. Interestingly, Maxwell deduced that the wave solutions to his equations (*in vacuo*) should move at a velocity

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = 3 \times 10^8 \text{ m/s} . \quad (2.2.7)$$

This was a shocking realization for physicists of the time. 3×10^8 m/s had been experimentally determined as the speed at which light propagates in a vacuum. Could this be a coincidence? Maxwell was the first to make the claim, “we can scarcely avoid the inference that light consists in the transverse undulations of the same medium which is the cause of electric and magnetic phenomena.” The equivalence of EM waves and visible light was later strengthened by the experiments of Heinrich Hertz⁶. This was one of the first great unifications in physics – two seemingly independent fields, electromagnetism and optics, are different manifestations of the same underlying physical principles – Maxwell's equations.

This is our first encounter with radiation. **Radiation** is the transfer of energy from one location to another by particles and waves. Here, we've described electromagnetic radiation. If

⁶Hertz was able to create and detect EM waves with electric circuitry. He also observed reflection, refraction, and polarization of EM waves, which were known properties of visible light.

I move an electric charge from point A to point $A + 1$ m here on Earth, my action on the charge causes a ripple in electric and magnetic fields. That ripple propagates out to spatial infinity. If those ripples travel out as EM waves to Alpha Centauri, where they in turn shake some other electric charges, then I have done work (caused a change in energy) on the charges at Alpha Centauri using my charge here. Electromagnetism, in this case, has mediated a transfer of energy and momentum.

There are a couple properties of EM waves which tend to show up on radiological physics exams. The first being calculations involving frequency and wavelength. For a propagating wave moving with velocity c , **wavelength** λ , and **frequency** (cycles per unit time) ν , the three quantities are related by

$$c = \lambda \nu . \quad (2.2.8)$$

For example, if an EM wave with wavelength $\lambda = 3 \cdot 10^{-5}$ m, its frequency is uniquely determined by

$$\nu = \frac{c}{\lambda} = \frac{3 \cdot 10^8}{3 \cdot 10^{-5}} = 10^{13} \text{ s}^{-1} = 10^{13} \text{ Hz} \quad (2.2.9)$$

THE ELECTROMAGNETIC SPECTRUM

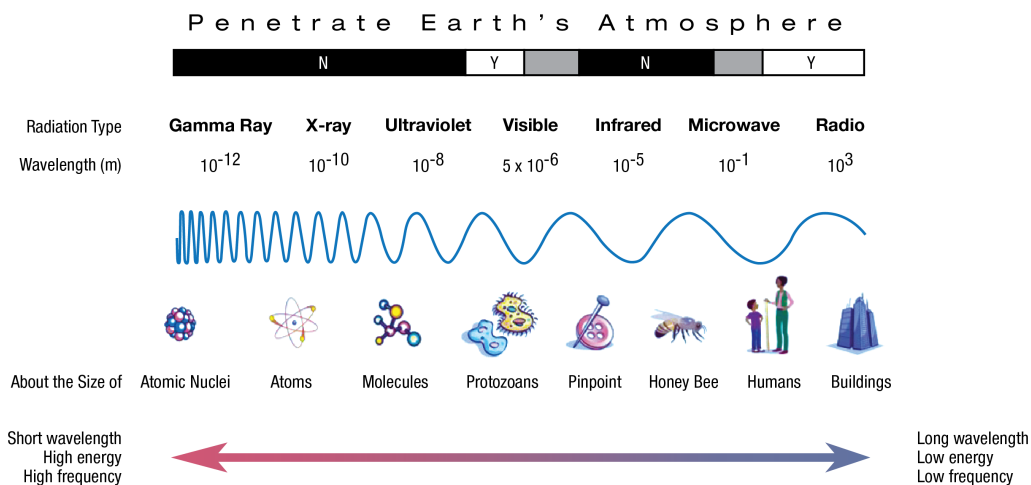


Figure 2.6: Electromagnetic waves are categorized by their wavelength. Different wavelengths of EM radiation are given different names.

One last topic we should discuss in electromagnetism before moving on is the definition of a commonly used unit of energy. Energy, in classical macroscopic phenomena, is usually

measured in the S.I. unit of Joules, where

$$1 \text{ J} = 1 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} . \quad (2.2.10)$$

In electromagnetism, energy is imparted by fields, with the amount of energy transferred being proportional to the field strength and the charge magnitude. Here, the Joule has an alternative definition as the amount of energy gained by 1 C of charge traveling 1 m in an electric field of strength 1 N/C,⁷

$$1 \text{ J} = 1 \text{ C} \frac{\text{N}}{\text{C}} \text{m} = 1 \text{ N} \cdot \text{m} = 1 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} \quad (2.2.11)$$

This unit of energy is fine for describing physics of our everyday lives, but is inadequate for discussing physics on microscopic scales. Instead, we often use the **electron-volt** eV. 1 eV is defined similarly to equation 2.2.11, but specifies the amount of energy gained by moving a single electron of charge $1.602 \cdot 10^{-19} \text{ C}$ by 1 m across an electric field of strength 1 N/C.

$$1 \text{ eV} = (1.602 \cdot 10^{-19} \text{ C}) \cdot 1 \frac{\text{N}}{\text{C}} \text{m} = 1.602 \cdot 10^{-19} \text{ J} \quad (2.2.12)$$

The electron-volt is often more practical for our discussions.

2.3 Relativity

2.3.1 The Michelson-Morley Experiment

The equivalence of light and electromagnetic waves came to be widely believed by physicists in the late 19th century. Familiar waves at the time, such as the mechanical waves created by dropping a stone in a lake, were known to propagate through a medium. In the stone-lake example, the disturbance in water caused by the stone is transferred through the medium of water. It was natural to think that light waves were therefore ripples in some medium which was sitting like a backdrop in space with celestial bodies moving within it. This theorized medium was called the “luminiferous ether”.

If this was the case, conventional wisdom⁸ would suggest that EM waves created on Earth, would gain or lose some speed depending on how we’re moving through the ether. This

⁷ $T = qEd$, where T is the charge’s energy, q is the charge magnitude, E is the (constant) electric field strength, and d is the distance traveled.

⁸Formally, Galilean Relativity.

is in the same way that, if I'm standing on a train moving with respect to the ground with speed v and I throw a baseball in the same direction that the train is moving with speed u with respect to the train, the ball then moves at speed $v + u$ with respect to the ground. Light, depending on the direction of its motion with respect to the ether, was expected to travel at different velocities (Figure 2.7).

Two physicists, Albert Michelson and Edward Morley, devised an experiment to determine the velocity of Earth within the ether using the relationships shown in Figure 2.7. They created a device called an interferometer (Figure 2.8), which was meant to measure the difference in time that it takes for two perpendicular rays of light to travel the same distance. Interestingly, this experiment was repeated several times without any significant difference in time for light to travel both distances.

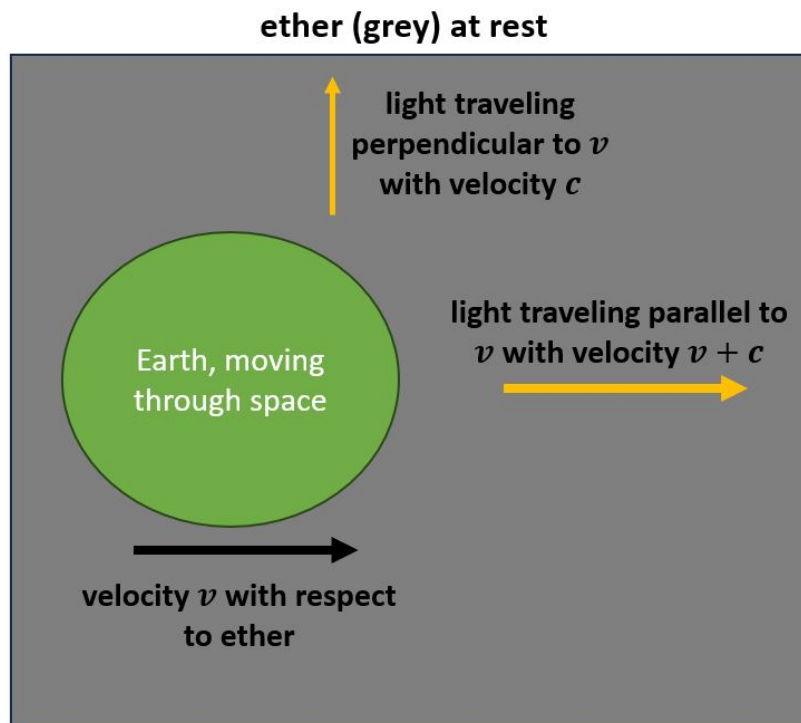


Figure 2.7: According to the ether hypothesis, light travels with speed $c + v$ depending on the motion of its origin within the ether.

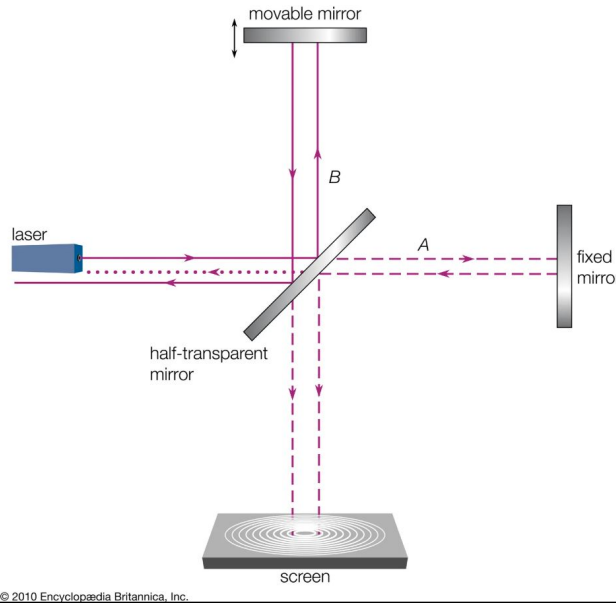


Figure 2.8: The interferometer devised by Michelson and Morley was meant to measure a difference in time that it takes for two perpendicular rays of light to travel the same distance. Their null result implies the constancy of the speed of light, from which Einstein deduced Special Relativity.

2.3.2 Einstein’s Special Theory

The null result of the Michelson-Morley experiment puzzled physicists for roughly two decades. However, in 1905, Albert Einstein made sense of this result – that light always travels at the same speed $c = 299792458$ m/s – with his Special⁹ Theory of Relativity. All of the consequences of Einstein’s Special Theory can be deduced from two postulates, namely

1. The laws of physics are identical for all observers in uniform motion.
2. The speed of light is a constant.

The above two statements imply lots of interesting consequences. First, if you and I are moving with respect to one another, but we both measure light as moving with the same speed (distance per time), something *must* be affecting our notions of time and/or space.

⁹The “special” theory which Einstein published in 1905 refers to the special case where observers are taken to move in uniform motion, not accelerating with respect to one another. The main results of the general case where subjects are allowed to accelerate, called “General” Relativity, were published by Einstein in 1915.

The specifics of how time and space, or the intertwined entity “spacetime”, is affected when different observers are moving with respect to one another can be deduced from the postulates above and a little algebra. We’ll skip straight to the result, where we find that relative distances and times between observers moving with respect to one another are related through a scalar quantity called the **Lorentz factor**

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} . \quad (2.3.1)$$

If we have coordinate systems S and S' which are coincident with one another at time $t = t' = 0$, and the frame S' is moving with velocity v in the $\hat{\mathbf{x}} = \hat{\mathbf{x}}'$ direction (Figure 2.9, lengths Δx and times Δt as measured in S' are related to lengths and durations are measured in S by

$$\Delta x' = \frac{1}{\gamma} \Delta x \quad (2.3.2)$$

$$\Delta t' = \gamma \Delta t \quad (2.3.3)$$

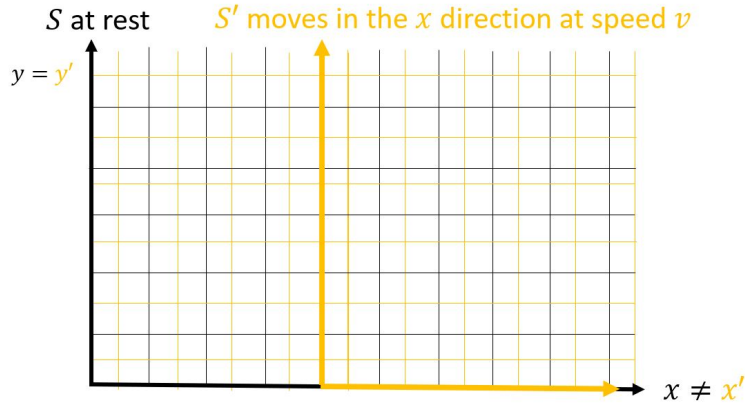


Figure 2.9: The Special Theory of Relativity describes coordinate systems moving with respect to one another with constant velocity v .

These phenomena are called **length contraction** and **time dilation**. If a massive (just meaning $m \neq 0$) particle is ejected from the sun in a solar flare towards the earth at 99.99% of the speed of light, $0.9999c$, and travels a distance of $d = 1.49 \cdot 10^{11}$ m to the Earth, a duration of

$$\Delta t = \frac{d}{c} = \frac{1.49 \cdot 10^{11}}{0.9999 \cdot 3 \cdot 10^8} = 500 \text{ s}$$

will have passed using our stationary clocks here on earth. However, in the coordinate system where the particle is at rest (our 500 seconds measured in the moving coordinates), the particle only experiences a duration of

$$\begin{aligned}\Delta t &= \Delta t' \frac{1}{\gamma} \\ &= \Delta t' \sqrt{1 - \frac{v^2}{c^2}} \\ &= (500 \text{ s})(\sqrt{1 - 0.9999^2}) \\ &\approx 7 \text{ s} .\end{aligned}$$

The statement that spacetime, and time specifically, has no absolute definition that we all agree upon ($t' \neq t$) marks a radical departure from the previous theories we've discussed and from intuition. Nevertheless, predictions from Einstein's Special Theory, and later his General Theory, hold up against experiment. In the famous Hafele-Keating experiment, for example, kinematic time dilation as described in equation 2.3.2 was confirmed by taking atomic clocks aboard airplanes for extended flights.

Another important consequence from the 1905 Theory of Relativity is an update to our expressions of energy and momentum. The Newtonian expression for kinetic energy,

$$T = \frac{1}{2}mv^2 , \tag{2.3.4}$$

does not comply with our second postulate of relativity, that the speed of light is constant. If I point my rocket ship at Andromeda and begin to burn through more and more fuel, I am transferring more and more chemical potential energy into the kinetic energy of the ship. There is nothing in our framework which limits the amount of kinetic energy we can transfer to an rocket ship, but there is a limitation that $v < c$. Our expression 2.3.4 clearly needs revising.

The correct revision to equation 2.3.4 accounts for an object's increased inertia – resistance to acceleration – as it gains kinetic energy. As a rocket ship approaches the speed of light, burning the same amount of fuel (or adding the same amount of energy) causes smaller and smaller increases in the ship's velocity such that the ship approaches but never reaches

c. The mass of the ship at rest, its **rest mass** m_0 , is replaced with a relativistic mass which accounts for its resistance to acceleration $m_0 \rightarrow m = \gamma m_0$. In this framework, the total energy of a body in motion is given by

$$E = \gamma m_0 c^2 \quad (2.3.5)$$

This is equal to the sum of the body's **mass energy**, its inherent energy which was required to create mass out of the void,

$$E_m = m_0 c^2, \quad (2.3.6)$$

and its **relativistic kinetic energy**, the *vis viva* of a body in motion,

$$T = m_0 c^2 (\gamma - 1). \quad (2.3.7)$$

Einstein's equations, specifically 2.3.6, tell us that mass and energy are different manifestations of the same thing, related by the constant c . This equivalence is critical to understanding radioactive decay, pair production, and other systems where matter is created or destroyed. What is striking is the amount of energy contained in even minuscule amounts of mass. If Marvin the Martian takes 1 kg of moon rocks (or 1 kg of moon feathers) and puts it into his atom vaporizer, he yields a sum of

$$\begin{aligned} E &= (1 \text{ kg})(3 \cdot 10^8 \text{ m/s})^2 \\ &= 9 \cdot 10^{16} \text{ J} \end{aligned}$$

of energy. To put this in perspective, NYC, consumes roughly $P = 5 \cdot 10^9 \text{ J/s}$. This means that the mass energy contained in 1 kg of matter could feed NYC's energy consumption for

$$\begin{aligned} \frac{m_0 c^2}{P} &= \frac{9 \cdot 10^{16} \text{ J}}{5 \cdot 10^9 \text{ J/s}} \\ &= 1.8 \cdot 10^7 \text{ s} \\ &\approx 208 \text{ days} \end{aligned}$$

Often, you'll find that people refer to particles' masses in terms of their mass energy. For example, we often say an electron's mass is

$$\begin{aligned} E &= (9.11 \cdot 10^{-31} \text{ kg})(3 \cdot 10^8 \text{ m/s})^2 \\ &= 511.0 \text{ keV} \end{aligned}$$

Similarly, a proton's rest mass is 938.3 MeV and a neutron's mass is 939.5 MeV.

2.4 Particle Physics

“What are we made of?” is one of the oldest questions in science and philosophy. The atomic hypothesis suggested by Greek philosopher Democritus in roughly 450 B.C.E. proposes that matter, on the smallest scales, is composed of indivisible building blocks called **atoms**, from the Greek *ατομος* meaning “uncuttable”.

Democritus claimed that our experience could be understood in terms of indivisible particles moving around in otherwise empty space. He claimed that atoms of different shapes and sizes underwent dynamics following deterministic laws: they could bounce off each other or stick together to make composite objects. Democritus also argued that many properties are not inherent to the atoms, but arise from their interactions. Wood atoms, for example, cannot be intrinsically brown since they are burned to create white ash.

Democritus had little experimental evidence to support the atomic hypothesis. One of his main arguments pertains to cutting an apple, supposing that the apple is not made of atoms, but some other continuous substance. If you drive a knife into the apple, the parts on either side of the knife get pushed away from the blade, but what happens to the stuff which is exactly on the boundary of the knife? Democritus concluded that there must be some empty space between apple atoms which are parted by the knife’s edge.

Substantial experimental evidence in support of the particle theory of matter didn’t come until roughly 2000 years after Democritus. In 1803, chemist John Dalton demonstrated the “law of multiple proportions” in mixtures of gasses. He observed, for example, carbon dioxide CO_2 contains roughly 2.66 g of oxygen per gram of carbon, while carbon monoxide CO contains 1.33 g of oxygen per gram of carbon. A different gas is formed when exactly twice the amount of oxygen is added to a mixture of CO . You don’t, for example,

- Yield carbon monoxide when carbon and oxygen are combined in equal amounts,
- Yield carbon dioxide when there is twice as much oxygen as carbon,
- Yield ammonia when there is π times as much oxygen as there is carbon.

Ethanol ($\text{C}_2\text{H}_6\text{O}$), lactic acid ($\text{C}_3\text{H}_6\text{O}_3$), glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), and formaldehyde (CH_2O) are not their own unique, fundamental substances. Rather, they all arise as combinations of carbon, hydrogen, and oxygen, in ratios of whole numbers.

Well – if ethanol, lactic acid, glucose, and formaldehyde, are not fundamental substances, what are? Carbon, hydrogen, and oxygen? Which properties distinguish hydrogen and oxygen on an atomic scale? Dmitri Mendeleev was the first to meaningfully organize experimental data regarding the 56 atomic elements which were known in 1867. He created the first periodic table (Figure 2.10), which sorted the elements in columns of similar chemical properties. Elements are also ordered from top to bottom and left to right by **atomic weight**.

Atomic weights A_w specify the amount of mass per atom of a given element. By convention, folks often use **atomic mass units (amu)**, defined as 1/12th of the mass of a carbon-12¹⁰ atom, which is approximately the mass of a hydrogen nucleus. Alternatively,

$$\begin{aligned} 1 \text{ amu} &= 1.66 \cdot 10^{-27} \text{ kg} \\ &= 931.5 \text{ MeV}/c^2 \end{aligned}$$

The amu, while great for discussing nuclear processes, is often impractical for our everyday purposes. Commercial airlines, for example, do not specify a checked-bag weight limit of $1.36 \cdot 10^{28}$ amu. To avoid this issue, chemists define a new unit called the **mole**

$$1 \text{ mol} = 6.022 \cdot 10^{23} \text{ atoms, molecules, units, etc.}$$

which is the amount of atoms contained in 12 grams of carbon-12. The atomic weights listed below each element on the periodic table can therefore be read as amu, or equivalently, g/mol.

$$\begin{aligned} 1 \frac{\text{amu}}{\text{atom}} \cdot 1 \text{ mol} &= (6.022 \cdot 10^{23} \text{ amu})(1.66 \cdot 10^{-24} \frac{\text{g}}{\text{amu}}) = 1 \text{ g} \\ &\rightarrow 1 \frac{\text{amu}}{\text{atom}} = 1 \frac{\text{g}}{\text{mol}} \end{aligned}$$

The number of particles within one mole of any substance is given the name **Avogadro's Constant** after the chemist Amedeo Avogadro. It is usually abbreviated N_A . If we have,

¹⁰The “12” in “carbon-12” indicates a nucleus containing 6 protons and 6 neutrons, although carbon, or things that have identical behavior to carbon, can exist in other forms, e.g., carbon-13, which has 6 protons and 7 neutrons. More on this in the next lecture.

for example, helium with atomic weight $A_w = 4.0026$ amu, we can use Avogadro's constant to calculate its

$$\text{atoms/gram: } N_A/A_w = 1.505 \cdot 10^{23} \text{ g}$$

$$\text{grams/atom: } A_w/N_A = 6.646 \cdot 10^{-24} \text{ g}^{-1}$$

or, if we know that there are Z electrons per atom,

$$\text{electrons/gram: } \frac{N_A Z}{A_w} = 3.009 \cdot 10^{23} \text{ g}^{-1}$$

Periodic Table of the Elements																		18 VIII 8A																	
1 IA 1A		2 IIA 2A																13 IIIA 3A		14 IVA 4A		15 VA 5A		16 VIA 6A		17 VIIA 7A		18 VIII 8A							
1 H Hydrogen 1.008				3 Li Lithium 6.941		4 Be Beryllium 9.012																5 B Boron 10.811		6 C Carbon 12.011		7 N Nitrogen 14.007		8 O Oxygen 15.999		9 F Fluorine 18.998		10 Ne Neon 20.180			
11 Na Sodium 22.990		12 Mg Magnesium 24.305		3 IIIB 3B		4 IVB 4B		5 VB 5B		6 VIB 6B		7 VIIB 7B		8 VIII 8		9 VIII 9		10 VIII 10		11 IB 1B		12 IIB 2B		13 Al Aluminum 26.982		14 Si Silicon 28.086		15 P Phosphorus 30.974		16 S Sulfur 32.066		17 Cl Chlorine 35.453		18 Ar Argon 39.948	
19 K Potassium 39.098		20 Ca Calcium 40.078		21 Sc Scandium 44.956		22 Ti Titanium 47.88		23 V Vanadium 50.942		24 Cr Chromium 51.996		25 Mn Manganese 54.938		26 Fe Iron 55.933		27 Co Cobalt 58.933		28 Ni Nickel 58.693		29 Cu Copper 63.546		30 Zn Zinc 65.39		31 Ga Gallium 69.723		32 Ge Germanium 72.61		33 As Arsenic 74.922		34 Se Selenium 78.09		35 Br Bromine 79.904		36 Kr Krypton 84.80	
37 Rb Rubidium 84.468		38 Sr Strontium 87.62		39 Y Yttrium 88.906		40 Zr Zirconium 91.224		41 Nb Niobium 92.906		42 Mo Molybdenum 95.94		43 Tc Technetium 98.907		44 Ru Ruthenium 101.07		45 Rh Rhodium 102.906		46 Pd Palladium 106.42		47 Ag Silver 107.868		48 Cd Cadmium 112.411		49 In Indium 114.818		50 Sn Tin 118.71		51 Sb Antimony 121.760		52 Te Tellurium 127.6		53 I Iodine 126.904		54 Xe Xenon 131.29	
55 Cs Cesium 132.905		56 Ba Barium 137.327		57-71 Lanthanide Series		72 Hf Hafnium 178.49		73 Ta Tantalum 180.948		74 W Tungsten 183.85		75 Re Rhenium 186.207		76 Os Osmium 190.23		77 Ir Iridium 192.22		78 Pt Platinum 195.08		79 Au Gold 196.967		80 Hg Mercury 200.59		81 Tl Thallium 204.383		82 Pb Lead 207.2		83 Bi Bismuth 208.980		84 Po Polonium [209]		85 At Astatine [210]		86 Rn Radon 222.018	
87 Fr Francium 223.020		88 Ra Radium 226.025		89-103 Actinide Series		104 Rf Rutherfordium [261]		105 Db Dubnium [262]		106 Sg Seaborgium [266]		107 Bh Bohrium [264]		108 Hs Hassium [265]		109 Mt Meitnerium [268]		110 Ds Darmstadtium [271]		111 Rg Roentgenium [272]		112 Cn Copernicium [285]		113 Nh Nihonium [286]		114 Fl Flerovium [289]		115 Uup Ununpentium [291]		116 Lv Livermorium [293]		117 Uus Ununseptium [294]		118 Uuo Ununoctium [294]	
Lanthanide Series				57 La Lanthanum 138.906		58 Ce Cerium 140.115		59 Pr Praseodymium 140.908		60 Nd Neodymium 144.24		61 Pm Promethium 144.913		62 Sm Samarium 150.36		63 Eu Europium 151.965		64 Gd Gadolinium 157.25		65 Tb Terbium 158.925		66 Dy Dysprosium 162.50		67 Ho Holmium 164.930		68 Er Erbium 167.26		69 Tm Thulium 168.934		70 Yb Ytterbium 173.04		71 Lu Lutetium 174.967			
Actinide Series				89 Ac Actinium 227.028		90 Th Thorium 232.038		91 Pa Protactinium 231.036		92 U Uranium 238.029		93 Np Neptunium 237.048		94 Pu Plutonium 244.064		95 Am Americium 243.061		96 Cm Curium 247.070		97 Bk Berkelium 247.070		98 Cf Californium 251.080		99 Es Einsteinium [252]		100 Fm Fermium 257.095		101 Md Mendelevium 258.1		102 No Nobelium 259.101		103 Lr Lawrencium [262]			

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Figure 2.10: Mendeleev identified the periodicity of the elements, organizing them in columns according to their similar chemical properties and their masses. His original table included 56 elements.

Mendeleev and other chemists, periodic table in hand, observed that all elemental weights appeared to be near multiples of the mass of the hydrogen atom. Surely it isn't a coincidence that helium weighs almost exactly twice as much as hydrogen, lithium three times as much, and so on, at least for the first few rows of elements. It's natural to wonder if hydrogen is to other atoms what atoms are to molecules. In the same way that integer ratios of elements combining to form chemicals hints at the existence of atoms, atomic masses being near integer multiples of the mass of hydrogen hints that the elements might be composed of more fundamental, constituent particles.

The internal structure of the atom was probed by experiments which are largely beyond the scope of these lectures. Different models of the atom – solid spheres, plum pudding, planetary orbits – came and went as experiments ruled out certain arrangements (Figure 2.11).

- In 1897, J.J. Thompson stripped gasses of their electrons, identifying electrically neutral atoms as being composed of positive and negative electric charges. He identified electrons as being relatively light ($m_e \approx \frac{1}{1800} m_p$ where m_e and m_p are the mass of the electron and proton, respectively) in comparison to atomic weights.
- In 1911, Ernest Rutherford bombarded gold foil with helium nuclei and, based on the distribution of the scattered helium, identified the positive charges within an atom as being part of a dense atomic nucleus. He called these particles protons. Based on this experiment, Rutherford also calculated the size of the nucleus $\sim 10^{-15}$ m in relation to the size of the atom $\sim 10^{-10}$ m, “a fly in a cathedral”.
- In 1931, James Chadwick demonstrated the existence of neutrons, particles of roughly the same mass as protons but without electric charge, by measuring the momentum transferred by a neutron beam incident upon paraffin wax. In doing so, he discovered the final major piece of the atom.

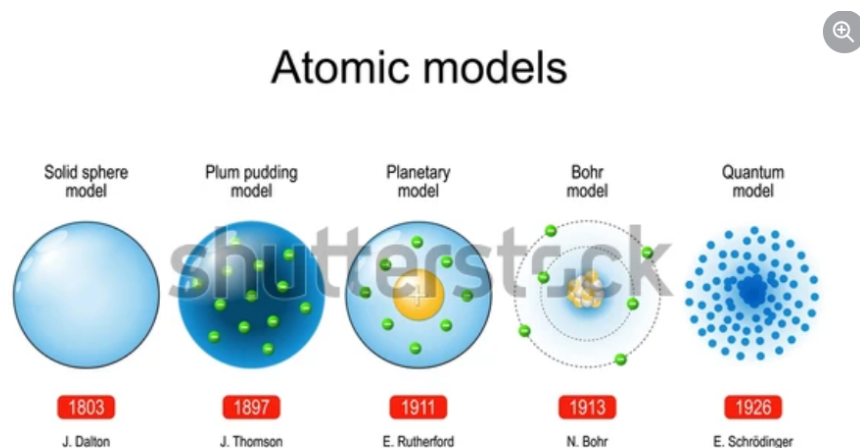


Figure 2.11: Historical development of atomic theory, courtesy of Shutterstock.

The atomic model of a nucleus densely packed the protons and neutrons surrounded by orbital electrons is not complete by modern standards, but we will not need much more to adequately describe radioactive decay.

2.4.1 Quantum Mechanics

In the beginning of the 1900's there were several experiments which suggested that classical laws of physics were inadequate to describe microscopic phenomena. These measurements were eventually resolved by the theory of **quantum mechanics (QM)**. The word “quantum” is used to indicate that we’re talking about discrete lumps of energy, rather than a continuum. Particles in QM don’t come in thirds, fifths, or $\frac{1}{\sqrt{2}}$ ths; particles (and their bound energy states) exist in discrete levels or not at all. I’ve listed three major experiments which motivated the quantum theory below.

- **Blackbody radiation (1900):** Classical electromagnetism and thermodynamics predicts frequency of electromagnetic radiation emitted from an idealized opaque, non-reflective body, called a “blackbody”. The distribution of wavelengths emitted is predicted as a function of the body’s temperature. The classical theory agrees with experimental results for long wavelengths, but disagrees significantly for wavelengths at and below the ultraviolet ($\sim 1 \mu\text{m}$). This divergence is dramatically referred to as the “ultraviolet catastrophe” (Figure 2.12). Max Planck resolved the disagreement by postulating that energy from electromagnetic radiation is emitted from the blackbody in discrete “quanta” with energy $E = h\nu$ where $h = 6.626 \cdot 10^{-34} \text{ J}\cdot\text{s}$ is called **Planck’s Constant**. This suggests that light, in some circumstances, behaves like a particle, the *photon*.

I should make a quick point of saying that this is another common radiation physics exam question: given an EM wave’s wavelength λ or frequency ν , calculate it’s energy according to the quantum theory. So, if we have a photon of $\lambda = 30 \text{ nm}$, it’s frequency is

$$\nu = \frac{c}{\lambda} = \frac{3 \cdot 10^8 \text{ m/s}}{30 \cdot 10^{-9} \text{ m}} = 10^{16} \text{ Hz} .$$

The energy of a photon with $\nu = 10^{16}$ Hz is then

$$\begin{aligned} E &= h\nu \\ &= (6.626 \cdot 10^{-34} \text{ J} \cdot \text{s})(10^{16} \text{ Hz}) \\ &= 6.626 \cdot 10^{-18} \text{ J} \end{aligned}$$

or,

$$\begin{aligned} E &= \frac{(6.626 \cdot 10^{-18} \text{ J})}{(1.602 \cdot 10^{-19} \text{ J/eV})} \\ &= 41.4 \text{ eV} \end{aligned}$$

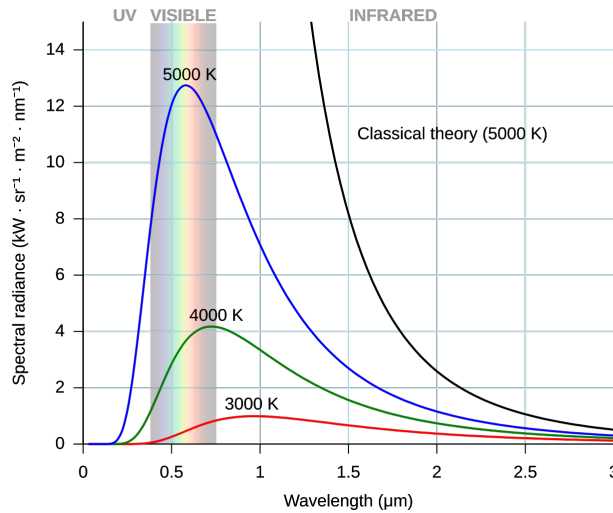


Figure 2.12: The predicted distribution of wavelengths according to classical theory (black) versus experimental results for blackbodies of various temperatures.

- **Photoelectric Effect (1905):** Electromagnetic waves, as described by Maxwell's equations, were known to carry energy and momentum. In the experiment shown in Figure 2.13, conducting metal plates are placed in a vacuum tube with a battery connecting the two plates. An electric field is present, but no electric current is measured by the ammeter in the unperturbed system. It had been shown that if EM waves above a certain frequency were shone onto one of the plates, the system would conduct current. Interestingly, it doesn't matter how *much* light is incident on the surface, just that the

light is of a certain color. For example, the circuit might not conduct a current if a bright red flood lamp was placed over the plate, but would conduct if a dim blue light was held over it. This was explained in an analysis by Einstein, where he again supposed that EM waves sometimes act like photons with energy $E = h\nu$, and that the photons only have the capacity to free atomic electrons (and conduct electricity) if they are above a certain energy.

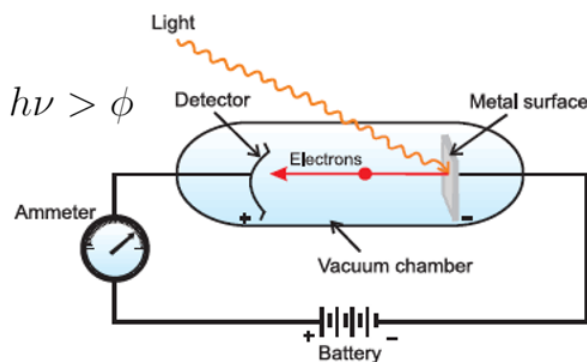


Figure 2.13: An experimental setup to observe the photoelectric effect. Light is shone onto a conducting electrode, but only creates a current if the photon energy $h\nu$ is greater than a threshold ϕ , specific to the metal.

- **Bohr Atom (1913):** If a sample of gas is brought to a high temperature, it will radiate energy away as electromagnetic waves. That light can be filtered through a prism to investigate the different wavelengths being produced by the sample (Figure 2.14). If you perform this experiment with hydrogen gas, you measure EM waves with specific wavelengths of 121.6 nm, 102.6 nm, 97.2 nm, and so on.

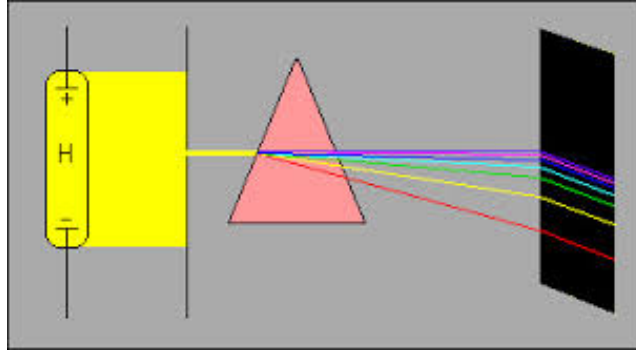


Figure 2.14: Experimental setup for emission spectroscopy using hydrogen gas.

This is a bit puzzling, right? Why should a hot gas emit light with wavelengths of 121.6 nm, 102.6 nm, etc., specifically? Why not a continuum?

This was explained by Niels Bohr with a model that became known as the **Bohr Atom**. Bohr observed that the wavelengths of light within the emission spectrum of hydrogen were related by

$$\nu = R \left[\frac{1}{m^2} - \frac{1}{n^2} \right]$$

where R is a constant and n, m are arbitrary whole numbers. He suggested that, rather than electrons in orbit around an atomic nucleus at any given distance, only specific distances/energy levels (electron orbitals) were permissible. The photons emitted by a heated gas then correspond to atomic electrons transitioning between discrete energy levels with $E = \frac{E_0}{n^2}$.

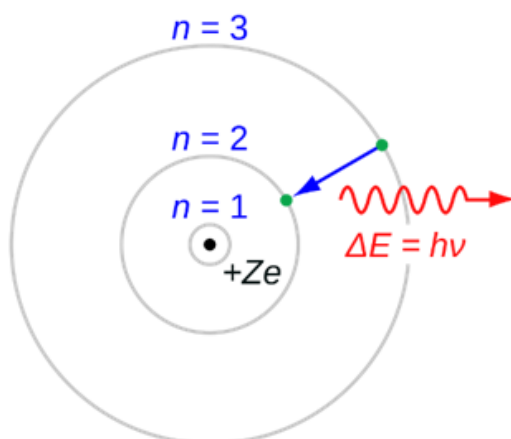


Figure 2.15: Niels Bohr's model of the atom. Photons of energy $h\nu$ are emitted during electron transitions between energy levels $\frac{E_0}{n^2}$ for $n \in \mathbb{Z}$.

All of this experimental evidence demonstrated an inadequacy of the classical physics theories of Newton and Maxwell. Mathematical descriptions of quantum mechanics which agreed with experiment were not developed until the 1920's. Two very different, but eventually shown to be equivalent, formalisms were developed in parallel: the wave mechanics of Erwin Schrodinger and matrix mechanics of Werner Heisenberg. We won't get into the details of quantum theory in this course. However, we will run into some of its main tenets, reproduced below.

- **Indeterminism:** In classical mechanics, the dynamics of any system can be determined with perfect accuracy if enough information is provided. If I know the position and momentum of every atom at $t = 0$, I can calculate how the system evolves just using Newton's Laws and Maxwell's Equations. This is *not* true on the quantum scale. Classical determinism is replaced by probabilistic wave mechanics – there is a fundamental limitation on how well we can know some quantities.
- **Discretization:** As shown in the Bohr Atom, only certain bound state energies are permitted. In classical physics, we expect a continuum of possibilities. The Earth happens to be orbiting the sun at an average distance of 93 million miles, but there's no reason that it couldn't orbit at an average distance of 94 million miles, 95 million miles, or

anywhere in between. In quantum mechanics, often only a list of possible orbitals, or energies, or spin states, are possible.

- **Wave-particle Duality:** As shown by the blackbody spectrum and photoelectric experiment, classical EM waves sometimes behave like particles. Similarly, classical particles such as electrons sometimes exhibit properties of waves.
- **Spin:** Particles carry intrinsic angular momentum called **spin**. Charged particles with spin create small magnetic fields, like that of the Earth turning on its axis. These spins and the magnetic fields they create are why bar magnets maintain a constant magnetic field when not moving. Manipulating spin states is how we generate signal in magnetic resonance imaging.

2.4.2 Quantum Field Theory and the Standard Model

Ironically, the most successful theory of particle physics is not a theory of particles at all, but a theory of fields. The original theory of QM developed in the 1920's has a couple glaring problems for those who are familiar with special relativity. Ordinary QM gives time a special status as the dimension in which dynamics evolve. Einstein's theories, however, demonstrate that perceptions of time and space bend into each other depending on relative motion. A proper theory should put space and time on equal footing.

Quantum Field Theory (QFT) aims to reconcile the theories of ordinary quantum mechanics and special relativity. It does so by treating everything in the universe as *fields* from which particles emerge. Fields, like in classical electromagnetism, mediate interactions between particles in a way that's consistent with *locality*. Shaking an electron over here creates a disturbance in the electromagnetic field which then propagates to a positron over there. In QFT, however, the electron is treated as lump in the electron field, the positron is a lump in the positron field, and their interaction is mediated by photons – lumps in the photon field.

Remarkably everything that we experience in our daily lives can be simplified down to interactions between a handful of fundamental fields. The modern analog of the periodic table is shown in Figure 2.16. There are 12 matter fields (and antimatter fields) in total: all of the

purple (quarks) and green (leptons) particles. Most of the matter in our daily experience is made of up quarks, down quarks, and electrons. Up and down quarks make up protons (u, u, d) and neutrons (u, d, d). Neutrinos are very light, electrically neutral particles which are quite antisocial; they can travel through a light-year of lead with only a 50% change of interacting. We'll come across them again when we discuss radioactive decay.

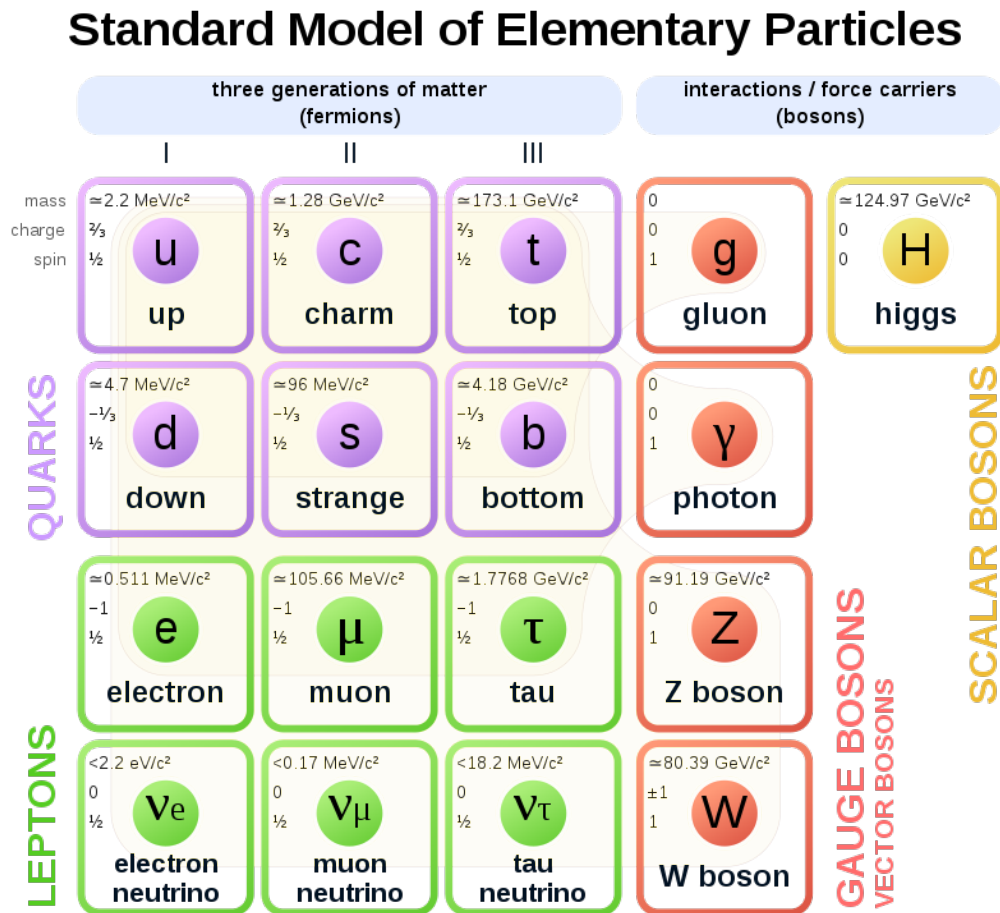


Figure 2.16: The modern version of the periodic table. The Standard Model describes 12 matter particles and their influence under the electromagnetic, nuclear strong, and nuclear weak forces. The Standard Model does not include a particle theory of gravity.

The matter particles have interactions which are mediated by the force carrier fields shown in red. Photons mediate the electromagnetic force, which we're already familiar with. Gluons mediate the strong nuclear force, which is responsible for binding together protons

and neutrons in the nucleus despite the enormous electrostatic repulsion of proton charges packed into 10^{-15} m. W and Z bosons mediate the weak force, which provides a mechanism for radioactive decay and nuclear fusion. The Higgs field, sometimes identified as nature's fifth force, gives the quarks, leptons, and weak bosons their mass.

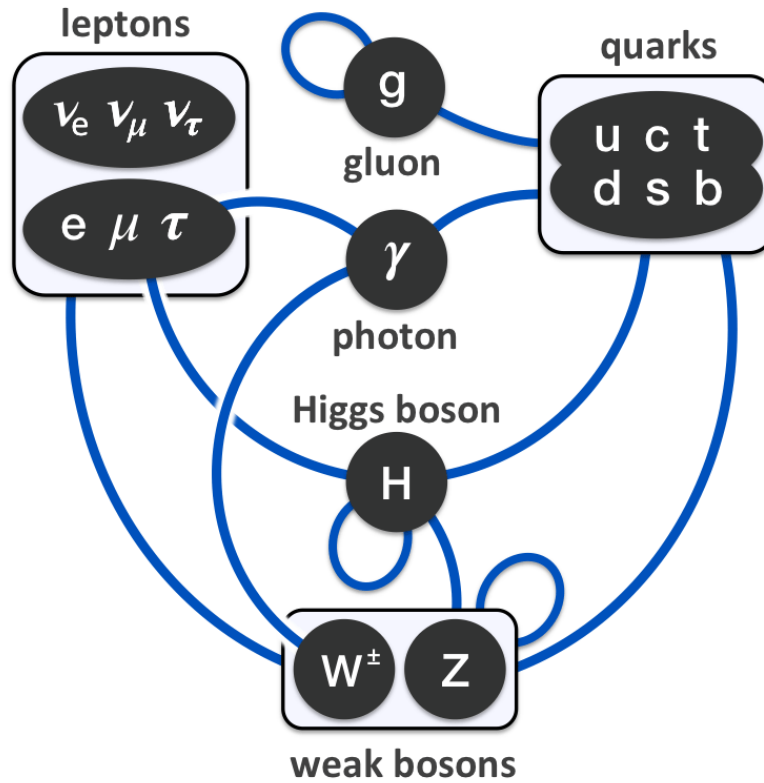


Figure 2.17: In the Standard Model, matter particles composed of quarks and leptons interact with each other through force carrier particles. Capacity for interaction between particles is depicted here with blue lines.

The Standard Model is one of the highest achievements of science. The QFT-predicted and experimental values of the magnetic moment of the electron – the strength of an individual electron's magnetic field – agree to about 1 part in 10^{10} . Still, there are many questions for which the Standard Model does not provide answers. The most obvious gap is a lack of a quantum description of gravity. Notably, astronomical and cosmological phenomena such as dark matter, too, lack explanations. Nevertheless, the Standard Model and QFT represent

how far we've come since Newton formulated his laws only 350 years ago.

Lecture 3

Radioactive Decay and Nuclear Reactions

3.1 Historical Overview

3.1.1 Röntgen and X-rays

At the beginning of 1896, Wilhelm Röntgen shocked the scientific community with his announcement of a “new kind of invisible ray,” *x-rays*, which he discovered while working with a Crookes tube. Crookes tubes are experimental apparatuses that were commonly used in the late 19th century to study the properties of electrons. In Crookes tubes, a large electric field is created between two conducting plates in a near-vacuum. If the voltage (electric field strength) between the two plates is sufficiently high, electrons are stripped from the small amount of air in the tube, and an electric current flows from the negatively charged plate, the cathode, to the positively charged plate, the anode, as a beam of electrons (Figure 3.1).

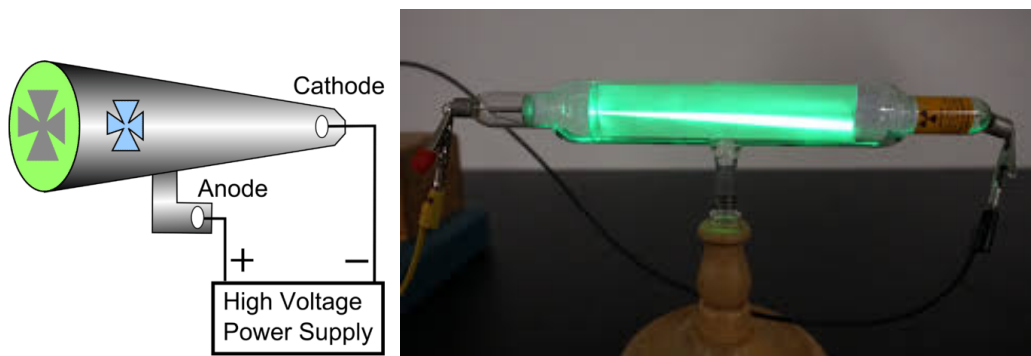


Figure 3.1: Schematic and demonstration of a Crookes tube.

In his lab, Röntgen happened to have cardboard screens painted with a fluorescent substance called barium platinocyanide. Fluorescence and phosphorescence are properties of materials called phosphors. **Phosphors** are substances which absorb energy from light of one wavelength and release that energy as light of different wavelength within the visible spectrum.

Fluorescence describes materials which release visible light immediately, e.g. anything that shines under a blacklight. **Phosphorescence** refers to materials which release energy as visible light over an extended duration, like “glow-in-the-dark” toys.

While Röntgen was working with his Crookes tube, he noticed that barium platinocyanide would fluoresce when placed next to the anode of his tube. Wondering whether this was due to visible light generated from the ionization of air in the tube (see the green glow in Figure 3.1), he designed a black cardboard cover to tightly wrap the device. Still, in complete darkness, Röntgen observed the barium platinocyanide fluorescing. Röntgen coined term “x”-rays to describe the invisible radiation, due to its ability to cast shadows (“ray”) on the fluorescent screens and its mysterious nature (“x”). 6 months after his initial announcement, Röntgen used x-rays to develop a radiograph of his wife’s hand, at which she very naturally remarked, “I have seen my death” (Figure 3.2).



Figure 3.2: The first ever radiograph, an image of Anna Bertha Röntgen’s hand.

Now we know now that there’s nothing mysterious about “x”-rays; they’re the exact same *stuff* as visible light – the electromagnetic waves predicted by James Clerk Maxwell. Our

eyes, however, evolved to process electromagnetic radiation from the sun with wavelengths from about 300 nm to 800 nm (Figure 3.3) rather than x-rays.

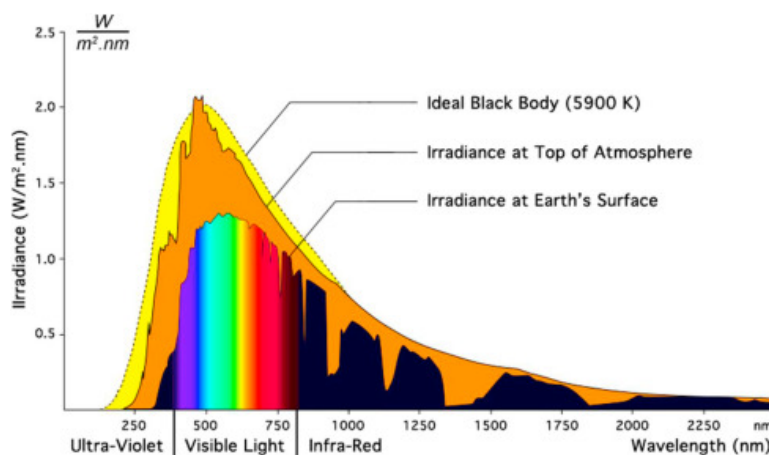


Figure 3.3: The most common EM wavelengths generated by the sun are those within the visible spectrum. <https://www.sciencedirect.com/topics/physics-and-astronomy/solar-spectra>.

3.1.2 Becquerel and Uranium Phosphorescence

Henri Becquerel, a French physicist, stumbled upon naturally occurring radioactivity in 1896. Becquerel had been studying phosphorescence of uranium salts, and in light of the news of Röntgen's discovery of x-rays, wanted to know if penetrating x-rays could be emitted from phosphorescent materials.

In his first few experiments, Becquerel would leave his uranium compounds out in the sun during the day, then wrap them in black paper so that no visible light would escape. He placed objects such as coins on top of a photographic plate, then placed his wrapped up uranium sample on top of the objects such that the shadow of the object would be projected in image on the developed plate. He initially concluded that x-rays were emitted as part of the phosphorescence process.

Becquerel continued on with his experiments, exposing the uranium and other materials to sunlight then developing photographic plates. The weather in Paris in late February of 1896, however, was overcast, and Becquerel could not continue exposing different phosphors to sunlight. Out of good intuition (or boredom), Becquerel chose to develop his photographic

plates anyway. To his surprise, an image still formed, just as dark as before, with a mark of a cross which had been placed between his uranium sample and the photographic plate (Figure 3.4). Becquerel made the first documented experimental observation of radioactive decay.

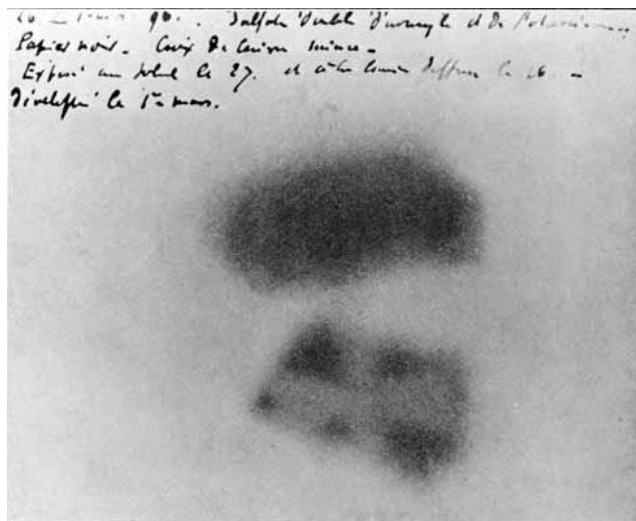


Figure 3.4: The photographic plate developed by Becquerel after not having exposed the uranium salts to sunlight.

Huh? Is this not a violation of energy conservation? If the uranium sample was not exposed to sunlight prior to being placed on the plate, where is the energy required to darken the film coming from (Figure 3.5)? We know now that Becquerel was observing alpha and beta decay within his sample, spontaneous changes in composition of the uranium nuclei to achieve a more energetically favorable configuration of the nucleus. Energy of **decay products** – what's left over after an instance of radioactive decay – comes from mass conversion of atomic nucleons.

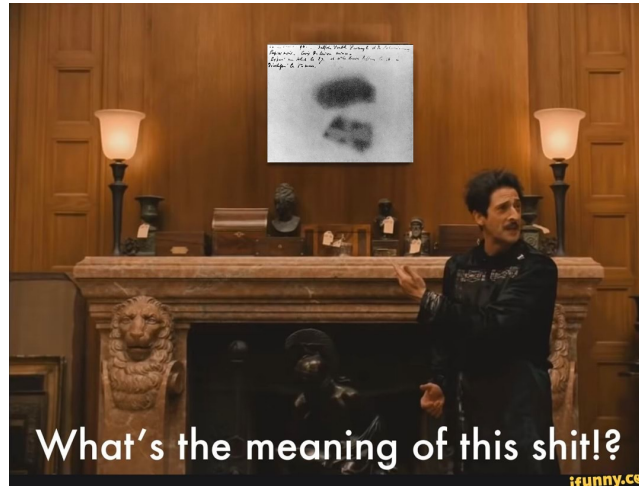


Figure 3.5: Does radioactivity violate conservation principles? (No.)

3.1.3 The Curies and Radium

While radioactive decay had been observed in uranium, “uranic rays”, it was unclear whether the phenomenon was specific to uranium or if it extended to other elements as well. Marie Curie had observed that uranium samples, when placed between two electrodes with a voltage applied between them, would cause the air in the chamber to conduct electricity.

This **ionization chamber** that she stumbled upon is one of the main instruments that we use to quantify radiation in the radiology and radiation oncology today. The principle of operation is that, when a moderate voltage is applied between two electrodes, the electrically neutral gas in the chamber gets split into electrically-not-neutral pieces which can conduct electricity. When energy is transferred to the gas by a radiation source, it kicks electrons out of their orbitals and creates positive and negative ions. Those ions then move through space like a current in a wire, with the amount of current being proportional to the number of electrons freed from their atomic orbitals.

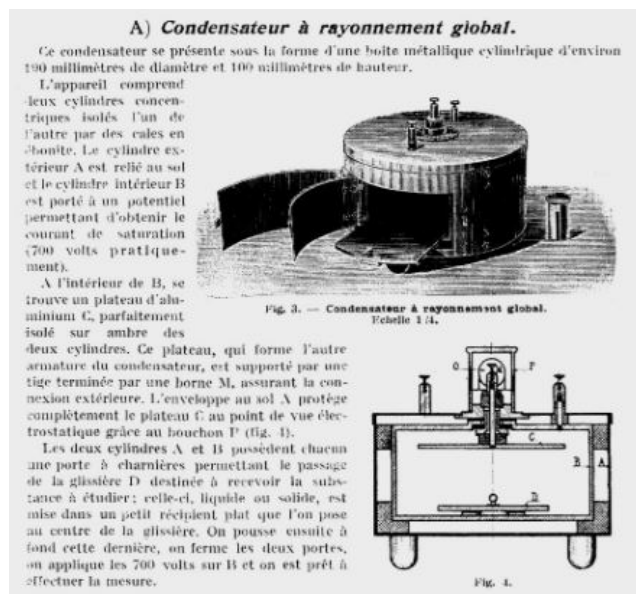


Figure 3.6: 1930 depiction of an ion chamber by a French instrument manufacturer, Charles Beaudouin.

Curie, with her ion chamber, set out on a systematic investigation to try to identify other radioactive substances. In doing so, she chemically isolated and identified radioactivity in two new elements, radium and polonium. To say that this process was tedious is an understatement. Curie, with the help of her husband Pierre, processed one ton of a uranium-rich ore called pitchblende in order to isolate just 0.1 g of RaCl_2 . Despite the tedium, the Curies were very prolific, having published about 30 scientific papers between 1898 and 1902. They are responsible for much of the formalism that I am presenting in the rest of this lecture.

3.2 Radioactive Decay

3.2.1 Atomic Physics Notation

Before we say any more, it will be helpful to describe some common nuclear physics notation. Particularly, it will be useful to define the atomic number Z and mass number A for different nuclei. Thanks to experimental results from the last lecture, we understand that atoms are composed of dense atomic nuclei containing relatively heavy, positively charged protons and neutral neutrons, while electrons carry a negative charge in a much larger volume in a quantum “cloud” around the nucleus.

Atoms are by default electrically neutral, with an equal number of protons and electrons. The number of protons in a nucleus, the **atomic number** Z , therefore determines the number of electrons in an atom assuming that the atom hasn't been **ionized** (stripped of some or all of its electrons). The electron cloud, because it inhabits most of the volume of an atom, talks with other atoms – creating and breaking chemical bonds. It's for this reason that an atom's electron orbitals are said to determine their chemical behavior. Because the number of protons and electrons determines the majority of our experience with the element, the atomic number is what we associate with an element's name. For example, we call all of the atoms with $Z = 2$ helium, $Z = 7$ nitrogen, $Z = 18$ argon, etc.

The atomic number doesn't describe everything about the nucleus. Atomic nuclei of the same element, defined by Z , might have different numbers of neutrons. If a nucleus has Z protons and N neutrons, it is said to have **mass number** $A = Z + N$. Chemical elements with the same number of protons and electrons, but different numbers of neutrons are called **isotopes**. Different isotopes of carbon, for example, might be indicated by their mass number: carbon-12 has 6 protons and 6 neutrons while carbon-13 has 6 protons and 7 neutrons. This is also indicated in shorthand with the following notation for arbitrary chemical X with Z protons and N neutrons,

$${}^A_ZX . \quad (3.2.1)$$

Or, again for carbon,

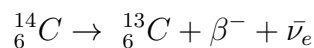
$${}^{12}_6C, {}^{13}_6C, {}^{14}_6C.$$

It's important to note that the atomic weights that are reported in the periodic table are weighted by the element's natural abundance. For example, chlorine primarily exists in nature as two isotopes, ${}^{35}\text{Cl}$ making up roughly 75% of all atoms and ${}^{37}\text{Cl}$ making up roughly 25%. If you look up the atomic weight of Cl on a periodic table, you'll find that that

$$\begin{aligned} A_{w,\text{Cl}} &= (0.75)A_{w,{}^{35}\text{Cl}} + (0.25)A_{w,{}^{37}\text{Cl}} \\ &= 0.75 \cdot 34.97 + 0.25 \cdot 36.97 \\ &= 35.47 \frac{\text{g}}{\text{mol}} \end{aligned} \quad (3.2.2)$$

If we're performing a calculation with an isolated sample of ^{37}Cl , it's appropriate to just use the mass number of the isotope for the calculation $A_{w,^{37}\text{Cl}} \approx 37$ amu.

When we discuss nuclear transformations, either from a nuclear reaction or a radioactive decay, we indicate the transformation process by an arrow, with the **reactants** on the left and the **reaction/decay products** on the right. For example,



3.2.2 Forces and Binding Energy

Of the four forces in nature (gravity, EM, strong, and weak), all but gravity are relevant for determining the dynamics of the nucleus. The electromagnetic force, mediated by photons, is present in the form of electrostatic repulsion between the protons with positive electric charges. The strong nuclear force, mediated by gluons, acts on the quarks within protons and neutrons to keep the nucleus tightly bound despite the electromagnetic force. The weak force, mediated by W and Z bosons, is responsible for some nuclear interactions and radioactive decay mechanisms.

You may have noticed in equation 3.2.2 that the atomic weights of chlorine-35 and chlorine-37 are neither 35 nor 37 amu. What gives? We defined 1 amu as 1/12th of the mass of carbon-12 atom. Maybe neutrons weigh slightly less than 1 amu and protons weigh slightly more than 1 amu, so the slight neutron-richness of the nucleus is driving the mass down? In fact, individual protons and neutrons while sitting out in space have a rest mass of slightly greater than 1 amu – 1.008 or so. When nucleons come together to form an atomic nucleus, they are entering a more energetically favorable arrangement. In order to do so in a way that conserves energy, they must trade off some of their mass in the form of binding energy. **Binding energy**, the amount of mass converted to energy when a nucleus is fused out of free protons and neutrons, tells us how favorable a particular nuclear arrangement is.

Consider the crude model of nuclear fusion shown in Figure 3.7. If two protons and two neutrons which are sitting free in space come together to form a helium nucleus, they are reducing their potential energy stored in the EM, strong, and weak force. That difference in energy is carried away from the nucleus as radiation. In stars, isotopes of hydrogen combine

to form more energetically favorable helium. That conversion of hydrogen mass into heat is the reason we have life on Earth.

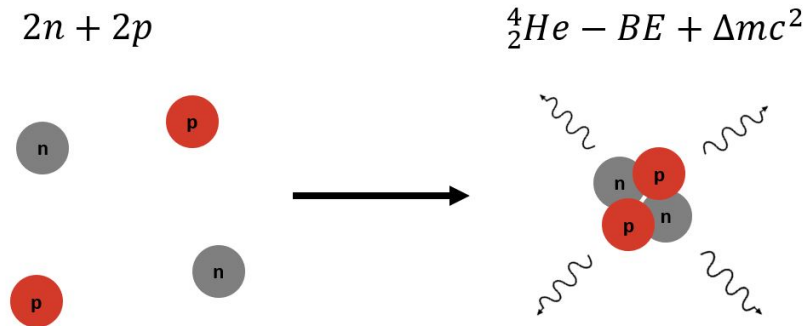


Figure 3.7: Unbound nucleons fuse together to form a helium nucleus. The difference in potential energy between their free and bound states $BE = \Delta mc^2$ is released as radiation.

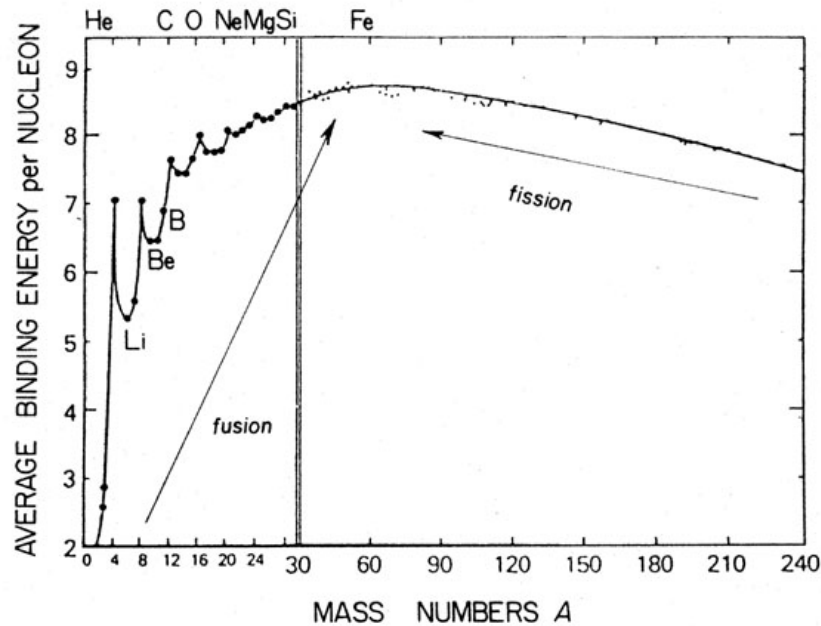


Figure 3.8: This plot of **binding energy per nucleon** as a function of mass number indicates which nuclei are more stable than others. Iron-56 is the most stable nucleus. Very light and very heavy nuclei have the potential to fuse and fission, respectively, in their aspirations to become more like iron. Y-axis units are arbitrary.

Regarding whether a nuclear arrangement is energetically favorable or favorable, it's similar how electrons are arranged in orbitals around the nucleus with the lower energy orbitals

filling up first. Atomic nucleons are arranged in shells, with some configurations of nucleons being more energetically favorable than others. The very-low-energy configurations occur when nuclear shells are completely filled, in the same way that noble gases are inert because their orbitals are perfectly filled. This occurs when Z or N are in the set of **magic numbers** $M = \{2, 8, 20, 28, 50, 82, 126\}$. Nuclei with both $Z \in M$ and $N \in M$ are called “doubly magic”, and are especially resistant to radioactive decay. The last doubly magic element in the periodic table is lead, ${}^{208}_{82}\text{Pb}$, beyond which every element is radioactive.

3.2.3 Radioactive Decay Mechanisms

In the same way that the gravitational force tells balls to roll down hills, EM and the strong and weak nuclear forces tell atomic nuclei to change their composition. Radioactive decay is a quantum mechanical process – it happens spontaneously with some probability of the decay happening at any given moment. There is nothing we can do predict when an individual atom is going to decay or to accelerate the rate at which a particular isotope decays. The probability of a nucleus undergoing a transition depends on how energetically unfavorable it is and how energetically favorable the target configuration is. There are a few different methods by which nuclei can change form which I’ll list here in Greek alphabetical order.

3.2.3.1 Alpha Decay

Alpha decay is prominent very heavy nuclei, meaning heavier than lead $Z = 82$. It occurs when electrostatic repulsion between protons overcomes the strong-force attraction between nucleons, and a particle containing 2 protons and 2 neutrons (doubly magic) is ejected from the nucleus. The particle, called alpha α , is identical to a helium ${}^4_2\text{He}$ nucleus. The decay is written for arbitrary chemical elements X and Y as

$${}^A_ZX \rightarrow {}^{A-4}_{Z-2}Y + \alpha + Q \quad (3.2.3)$$

Notice that the atomic number and mass number of Y are uniquely determined by the atomic number and mass number of X , since Y has exactly 2 protons and 2 neutrons less than X . Notice also that I’ve explicitly written the energy released in the decay process, symbolized by Q .

As an everyday example, naturally occurring radium-226 is abundant in soil in the United States; it decays to radon-222 gas which can build up in poorly ventilated spaces, causing unintended radiation exposure. This decay is written



In all decay processes, the energy Q comes from a difference in binding energy, or equivalently a difference in mass energy Δmc^2 . The difference in mass between the reactants and products of a nuclear reaction or decay is known as the **mass defect**. The decay shown in equation 3.2.4, for example, has

- Ra-226 $m_{\text{Ra}} = 226.025408$ amu,
- Rn-222 $m_{\text{Rn}} = 222.017570$ amu,
- Alpha $m_{\alpha} = 4.001506$ amu.

The difference in mass between the products and reactants is

$$\begin{aligned} \Delta m &= 226.025406 - (222.017574 + 4.002603) \\ &= 0.005229 \text{ amu} \end{aligned}$$

so the difference in energy is

$$\begin{aligned} Q &= \Delta mc^2 \\ &= (0.005229 \text{ amu})(931.5 \frac{\text{MeV}}{\text{amu}}) \\ &= 4.871 \text{ MeV} \end{aligned}$$

for every alpha decay of Ra-226. This energy is divided between the radium nucleus and the alpha particle. However, because *momentum* is also conserved in the decay, and the mass of radium nucleus is much greater than the alpha particle, the large majority of the decay energy is transferred to the alpha particle.

3.2.3.2 Beta Decay

While alpha decay occurs in nuclei which are simply too heavy, beta decay occurs in nuclei which have either too many neutrons or too many protons, and improve their stability by converting a neutron to a proton or vice versa. This interaction is mediated by the weak force. The process is shown (at first order) in Figure 3.9 if you're interested, but you don't need to be able to reproduce these diagrams.

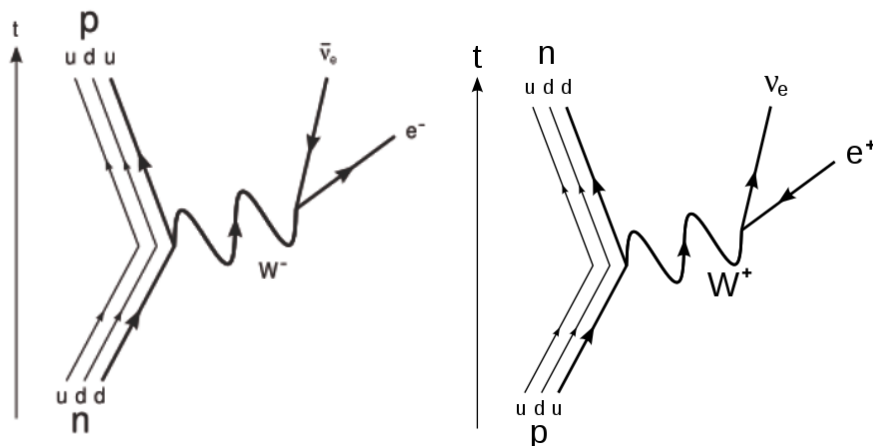


Figure 3.9: Beta-minus decay on the left and beta-plus decay on the right are mediated by W^- and W^+ bosons.

Beta-plus (positron) decay occurs in proton-rich nuclei which convert an atomic proton into a neutron like

$${}^A_Z X \rightarrow {}^A_{Z-1} Y^- + \beta^+ + \nu_e + Q \quad (3.2.5)$$

Similarly, beta-minus (negatron/electron) decay occurs in neutron-rich nuclei which convert and atomic neutron into a proton like

$${}^A_Z X \rightarrow {}^A_{Z+1} Y^+ + \beta^- + \bar{\nu}_e + Q \quad (3.2.6)$$

The ν_e and $\bar{\nu}_e$ here are indicating the neutrino and antineutrino shown in the diagram Figure 3.9. The energy in beta decay is split between beta particles and the neutrinos, such that the energy spectrum observed in beta decay is continuous.

Of note, the positron which is created in beta-plus decay does not get very far before it inevitably meets it's matter counterpart, and electron, and **annihilates**. When an electron

and a positron annihilate, their masses off 511 keV each are converted to two photons which travel in opposite directions. This is the principle behind positron emission tomography (PET), where a positron-emitting radio-tracer is ingested by the patient, then we set up our cameras and look for 511 keV photons.

3.2.3.3 Gamma Decay

In addition to “isotopes”, nuclei containing the same number of protons Z , physicists sometimes use the expressions

- **Isobar**: same atomic number A ,
- **Isotone**: same number of neutrons N ,
- **Isomer**: same Z , N , A , but different energy state.

Nuclei which are in less-than-optimal configurations can de-excite through *isomeric transitions*. When an isomer de-excites, it often releases a photon called a “gamma ray”. Gamma rays are physically identical to any other photon or EM wave of the same wavelength. Some people make the distinction that gamma rays come from nuclear transitions and x-rays come from bremsstrahlung or whatever else.

Isomeric transitions are the basis for single-photon emission tomography (SPECT) with technetium-99m, where the “m” stands for “metastable”. Tc-99m decays to Tc-99 and emits gamma rays. Those rays are then detected with a gamma camera. We can make write a reaction for gamma decays, it’s just pretty boring



3.2.3.4 Electron Capture

Electron capture occurs in proton-rich nuclei, similar to beta-plus decay. For this reason, it is said that the two mechanisms *compete* with each other. In electron capture, a proton in the nucleus is converted to a neutron, but it does so by absorbing an orbital electron, usually

from the K (lowest energy) shell.



The reaction will leave a vacancy in the K, L, or M, shell which then become occupied by an electron from a higher energy shell. Following electron capture or internal conversion, when the inner shell vacancy is filled, energy is released from the atom in one of two ways.

- **Fluorescent x-rays** might be emitted with unique energies corresponding to the differences in atomic energy levels.
- **Auger electrons** might be emitted. Rather than de-exciting by releasing a characteristic x-ray, an outer shell electron is sometimes launched from the atom when the inner shell is filled.

3.2.3.5 Internal Conversion

Internal conversion is a process that competes with gamma ray emission. When a nucleus is in an excited state, it might de-excite by releasing a gamma ray. Alternatively, it might kick an electron from an inner orbital. Internal conversion, like electron capture, leaves a vacancy in an inner shell. Following the ejection, the atom will de-excite through fluorescent x-ray or Auger electron emission.

3.2.4 Exponential Decay

The kinematics of radioactive decay are governed by the following differential equation

$$\frac{dN}{dt} = -\lambda N, \quad (3.2.9)$$

which states that, at any given moment the rate of change of atoms $\frac{dN}{dt}$ with respect to time t in a sample of radioactive material is proportional to the number of atoms in that sample N , with the constant of proportionality λ . If I have 200 atoms decaying at 2 atoms per second, when I have 100 atoms left, my sample will be decaying at 1 atom per second.

The solution to equation 3.2.9 is an exponential curve,

$$N(t) = N_0 e^{-\lambda t}, \quad (3.2.10)$$

where $N(t)$ is the number of atoms N in my sample as a function of time t , N_0 is the number of atoms I started with at $t = 0$, and λ is called the **decay constant** for a specific isotope. λ characterizes how quickly the sample of atoms is decaying away. Although the number of atoms is technically an integer $N \in \mathbb{Z}$, we're typically concerned with samples on the order of $N_A \approx 10^{23}$, so it's fair to just treat all of our variables in 3.2.10 as real numbers in $\mathbb{R}$.

When discussing radioactivity, we often specify the amount of radioactive stuff in terms of its **activity** α , which is just how many decays are occurring per unit time.

$$\alpha = -\frac{dN}{dt} = \lambda N \quad (3.2.11)$$

The S.I. units for activity are the **Becquerel** Bq, after Henri Becquerel, and indicates 1 decay per second. An often more practical unit is the **Curie** Ci, after Marie Curie, which is

$$1 \text{ Ci} = 3.7 \cdot 10^{10} \text{ Bq} . \quad (3.2.12)$$

A typical iridium-192 high dose rate brachytherapy source, for example, has around contains around 10 Ci of iridium. To avoid throwing around factors of 10^{10} , all the time, the Ci makes sense.

The decay constant, while it might be more natural for describing exponential decay, is not especially intuitive for describing how fast a sample is decaying. Rather than λ , we often use the half-life $T_{1/2}$: the amount of time it takes for half of a radioactive sample's atoms to decay. To find $T_{1/2}$ given λ , we plug in $N(t) = \frac{1}{2}N_0$ and solve for $T_{1/2}$.

$$\begin{aligned} \frac{1}{2}N_0 &= N_0 e^{-\lambda T_{1/2}} \\ \ln\left(\frac{1}{2}\right) &= -\lambda T_{1/2} \\ \rightarrow T_{1/2} &= \frac{\ln(2)}{\lambda} \end{aligned} \quad (3.2.13)$$

Another quantity of interest is the mean life. The mean life characterizes on average, how long will an atom from your sample of radioactive atoms live before it decays. Without going through any derivations, this quantity is

$$\tau = \frac{1}{\lambda} . \quad (3.2.14)$$

In isotope's mean life is useful for a few purposes in radiotherapy. For example, if I am performing low-dose-rate brachytherapy with a permanent prostate seed implant, the cumulative dose D to a point from the implant can be expressed as the product of the mean life τ and the initial dose rate to that point immediately following the procedure $\left(\frac{dD}{dt}\right)_{t=0}$,

$$D = \tau \left(\frac{dD}{dt} \right)_{t=0} . \quad (3.2.15)$$

If an I-125 implant with $\tau = 86$ days initially receives a dose rate of $\left(\frac{dD}{dt}\right)_{t=0} = 1.75$ Gy/day, the cumulative dose can be calculated

$$\begin{aligned} D &= \tau \left(\frac{dD}{dt} \right)_{t=0} \\ &= (86 \text{ days})(1.75 \text{ Gy/day}) \\ &= 150.5 \text{ Gy} . \end{aligned}$$

...

While you should always be able to work out solutions to radioactive decay problems using 3.2.10, there are a couple other forms of this equation which are convenient to know. First, because the activity of a sample and the number of atoms of a sample are related by equation 3.2.11, $\alpha(t) = \lambda N(t)$, we have

$$\alpha(t) = \alpha_0 e^{-\lambda t} \quad (3.2.16)$$

The activity of a sample decays exponentially with decay constant λ just like the number of atoms $N(t)$.

Sometimes it's also convenient to perform decay calculations with time expressed in units of half-lives. If a radioactive sample initially has N_0 atoms, after one half-life,

$$N(T_{1/2}) = N_0 \left(\frac{1}{2} \right)$$

after two half-lives,

$$N(2T_{1/2}) = N_0 \left(\frac{1}{2} \right) \left(\frac{1}{2} \right)$$

or after an arbitrary number n half-lives

$$\begin{aligned} N(nT_{1/2}) &= N_0 \left(\frac{1}{2}\right) \cdot \dots \cdot \left(\frac{1}{2}\right) \quad (n \text{ times}) \\ &= N_0 \left(\frac{1}{2}\right)^n . \end{aligned} \tag{3.2.17}$$

If I have a 10 Ci brachytherapy source of Ir-192 with a half-life of 74 days, and I want to know how much iridium remains after 111 days, or $111/74 = 1.5$ half-lives, I can use

$$\begin{aligned} \alpha(1.5 \text{ half-lives}) &= \alpha_0 \left(\frac{1}{2}\right)^{1.5} \\ &= (10 \text{ Ci})(0.354) \\ &= 3.54 \text{ Ci} . \end{aligned}$$

3.3 The Chart of the Nuclides

If we want to organize the elements including different combinations of possible atomic nuclei, the chemical periodic table created by Mendeleev in Figure 2.10 no longer does the trick. The updated catalog of the elements is called the **Chart of the Nuclides** and is shown in Figure 3.10. It includes all observed atomic nuclei, their masses, and their primary decay mechanisms. The color coding in Figure 3.10 is used to indicate a nucleus' stability quantified by its half-life. The stable elements which are not known to decay at all are shown in dark red, e.g. carbon-12. The elements which are very unstable are shown in light blue, e.g. carbon-20, which decays via β^- in $T_{1/2} \approx 16$ ms.

Notice that the stable elements do not fall along the line $N = Z$. While the same number of neutrons and protons makes for a stable nucleus in lighter elements, more neutrons are required for heavy elements to combat electrostatic repulsion between protons. The so-called **line of stability** is shown in an upward curve in Figure 3.10 with $N > Z$ for heavy nuclei. An interactive chart of the nuclides published by the International Atomic Energy Agency (IAEA) is also available at <https://www-nds.iaea.org/relnsd/vcharthtml/VChartHTML.html>.

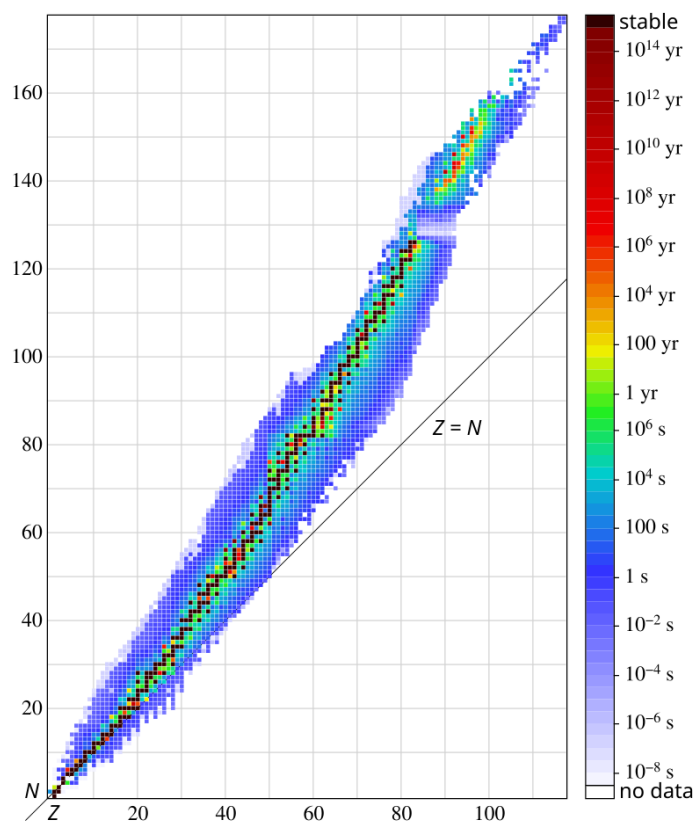


Figure 3.10: The complete chart of the nuclides with color coding to indicate stability of the nucleus, characterized by half-life.

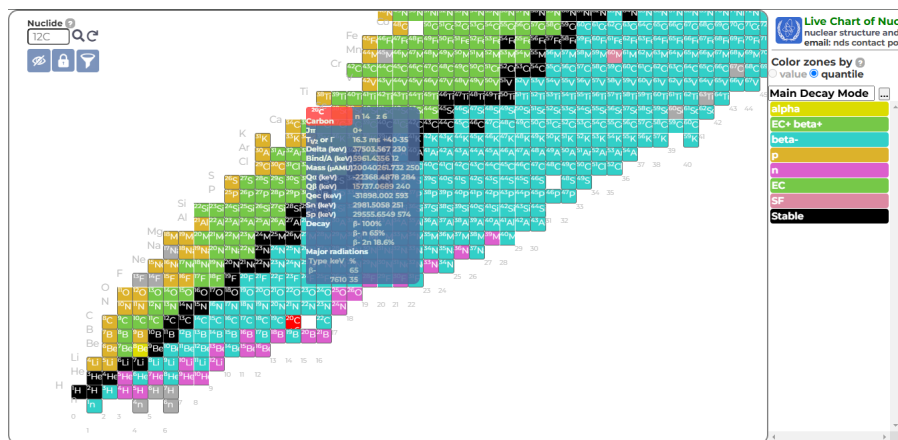


Figure 3.11: The IAEA interactive chart of the nuclides at <https://www-nds.iaea.org/relnsd/vcharthtml/VChartHTML.html>.

3.4 The Inverse Square Law

If there is a single topic that I can guarantee will be on the my exams, the ASRT registry, and pretty much any other radiation physics exam you'll come across, it's the **inverse square law**. The inverse square law states that a radiation source which behaves like a point **isotropically** (without preference for direction) emitting radiation, the intensity of the radiation decreases in proportion to the square of the distance from the source. Say we have some small sample of radioactive material which emits N particles per second as shown in Figure 3.12. Assuming the total number of particles is conserved, i.e. zero attenuation, the same number of particles N must be penetrating a sphere of radius $R = 1$ having surface area $4\pi(1)^2$, a sphere of radius $R = 2$ having surface area $4\pi(2)^2$, a sphere of radius $R = r$ having surface area $4\pi r^2$.

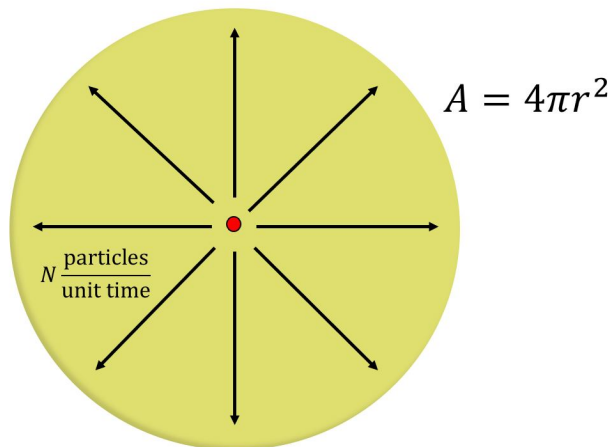


Figure 3.12: For a point source isotropically emitting N particles per unit time, the particles are evenly distributed over an area of $4\pi r^2$, where r is the distance from source.

The quantity we're describing here has a name – it's known as the particle **fluence rate** or **flux density**,

$$\dot{\phi} = \frac{dN}{dA \cdot dt} . \quad (3.4.1)$$

ϕ has dimensions of particles per area per time – it describes the number of particles dN passing through an area dA in the time interval dt .

$$[\dot{\phi}] = \frac{\text{particles}}{\text{cm}^2 \text{ s}}$$

In the case above of an unattenuated isotropic point source with activity $\alpha(t)$, the fluence rate¹¹ is a function of the distance r from the source and the time t that we look at the sample, $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}$. It is simply the source's activity divided by the surface area of a sphere with radius r ,

$$\dot{\phi}(r, t) = \frac{\alpha(t)}{4\pi r^2} . \quad (3.4.2)$$

The property that the fluence rate for a point source decreases with the square of the distance from the source,

$$\dot{\phi} \sim \frac{1}{r^2} , \quad (3.4.3)$$

has important implications for clinical practice. From a radiation safety standpoint,

Another term similar to fluence rate that we'll come across the aggregate particle **fluence**, which we'll call Λ . This is different from the fluence *rate*, and is a measure of the *total* number of particles dN that have passed through an area dA – we've summed over time. For example, if we have a radioactive source at $t = 0$ with N_0 particles, then have $N(t) = N_0 e^{-\lambda t}$ particles remaining after time t , the total number of particles which pass through a sphere of radius r is the difference between the initial sample and the remaining sample,

$$\phi(t) = \frac{N_0 - N_0 e^{-\lambda t}}{4\pi r^2} = \frac{N_0(1 - e^{-\lambda t})}{4\pi r^2} \quad (3.4.4)$$

The dimensions for ϕ are

$$[\phi] = \frac{\text{particles}}{\text{cm}^2} .$$

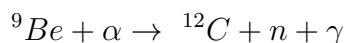
3.5 Nuclear Reactions

Alpha, beta, and gamma decay are sufficient to describe the majority of the phenomena we'll discuss in this class and encounter in the clinic. It should be stated, however, that these are not the only nuclear transformations which are possible. The machinery of the electromagnetic, strong, and weak forces accommodates other spontaneous decay mechanisms. For example, double beta decay, in which a two neutrons are spontaneously and simultaneously converted

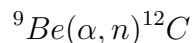
¹¹Also ignoring particle time of flight.

into two protons via the emission of two electrons and two electron antineutrinos, has been directly observed in over a dozen isotopes.

Beyond radioactive decay, in which atomic nuclei spontaneously disintegrate, nuclear reactions can be instigated by bombarding nuclei with external particle sources. For example, the experiment by which the neutron was discovered involved the bombardment of beryllium-9 with alpha particles. This process is depicted as



or with the shorthand



This is called an (α, n) , read “alpha-n”, reaction. In addition to (α, n) , reactions such as (α, p) , (p, γ) , (n, γ) , and (n, p) , are possible. These nuclear interactions are the means by which stable atomic nuclei might be transformed into unstable, radioactive species of atoms. This is referred to as **nuclear activation**. It is necessary to concern the activation of stable isotopes when nuclear interactions such as those listed above are likely, e.g., when designing shielding for a proton treatment facility.

One nuclear interaction that we are often concerned with in photon-based radiotherapy is referred to as **photodisintegration**, (γ, n) . In photodisintegration, a photon interacts with a nucleus and frees bound neutrons. This interaction is prominent in high- Z materials for photon beams with accelerating potentials greater than 10 MV. Clinical linear accelerators often shape the x-ray beam with tungsten, $Z = 74$. Photodisintegration interactions in the linac head between x-rays and tungsten are liable to create an undirected bath of neutrons within the treatment room. This is a concern for radiation safety as well as whole-body patient dose.

Finally, there are two methods of nuclear transformation which are not especially important for our purposes, but should be mentioned. **Nuclear fission** is a process in which very heavy nuclei spontaneously split into two lighter isotopes. Nuclear fission, usually of uranium or plutonium, is the mechanism underpinning modern nuclear power plants and the first atomic bombs. **Nuclear fusion** is the process in which two lighter species of nuclei combine

to form a more stable, heavier nucleus. Hydrogen combining to form helium is the process by which the sun “burns” and thermonuclear warheads operate.